Discrete Stability, Existence, and Convergence of a Fully Discrete FEM Scheme for Two-Phase Ferrohydrodynamics with the Full Shliomis Model

T1 (Landau–Lifshitz) + T2 (Spin–Vorticity) + T3 (Antisymmetric Stress)

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Abstract

We present a complete theoretical analysis of a fully discrete finite element scheme for two-phase ferrohydrodynamic (FHD) flows that includes *all three* nonlinear magnetization terms of the Shliomis model: the Landau–Lifshitz damping $-\beta\mathbf{M}\times(\mathbf{M}\times\mathbf{H})$ (T1), the spin–vorticity coupling $\frac{1}{2}(\nabla\times\mathbf{U})\times\mathbf{M}$ (T2), and the antisymmetric magnetic stress $\frac{\mu_0}{2}\nabla\times(\mathbf{M}\times\mathbf{H})$ (T3). The scheme is built on the energy-stable DG framework of Nochetto, Salgado, and Tomás [2] (which omits T1, T2, T3) and restores all three terms following Zhang, He, and Yang [3]. Our contribution is a single-step, non-decoupled, mixed CG–DG discretization for which we prove: (i) *unconditional* discrete energy stability, showing the β-term contributes the additional dissipation $+2\mu_0\beta\,\tau\,\|\mathbf{M}^{k-1}\times\mathbf{H}^k\|^2$ on the left-hand side of the energy law; (ii) existence and conditional uniqueness of the discrete solution at every time step (Brouwer fixed-point argument with two parabolic-type CFL conditions $\tau\le\tau_*\sim h^d/R^2$ for existence and $\tau<\tau_0\sim h^{2/3}/R^{2/3}$ for uniqueness); and (iii) weak convergence of subsequences to a weak solution of the continuous Shliomis system as $h,\tau\to 0$, under a uniform $L^4_{t,x}$ regularity assumption on the discrete magnetization.

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1 Introduction

We study a fully discrete finite element scheme for two-phase ferrohydrodynamic flows governed by the coupled Cahn–Hilliard / Navier–Stokes / magnetization system of Nochetto, Salgado, and Tomás [2], extended to include all three nonlinear terms dropped in the original work. The theoretical analysis is presented in Sections 4–6.

2 The Continuous Model

2.1 Physical Model and Governing Equations

We consider a two-phase ferrofluid system occupying a bounded, convex, polyhedral domain $\Omega \subset \mathbb{R}^d$ $(d = 2, 3)$, with boundary $\Gamma = \partial\Omega$ and final time $T_f > 0$. The model couples a diffuse-interface description of the fluid–fluid interface (Cahn–Hilliard), incompressible viscous flow (Navier–Stokes), magnetization dynamics (Shliomis), and magnetostatics (Poisson).

Phase field. A scalar order parameter $\theta : \Omega \times [0, T_f] \to \mathbb{R}$ distinguishes the two fluid phases ($\theta \approx \pm 1$). It evolves by the Cahn–Hilliard system:

$$\partial_t \theta + \mathbf{u} \cdot \nabla\theta = \gamma \Delta\psi, \qquad \psi = -\varepsilon\Delta\theta + \tfrac{1}{\varepsilon} f'(\theta), \tag{1}$$

where ψ is the chemical potential, $\varepsilon > 0$ is the interface thickness parameter, $\gamma > 0$ is the mobility, and $f(\theta) = \frac{1}{4}(\theta^2 - 1)^2$ is the double-well potential.

Fluid flow. The incompressible velocity $\mathbf{u} : \Omega \times [0, T_f] \to \mathbb{R}^d$ and pressure $p : \Omega \times [0, T_f] \to \mathbb{R}$ satisfy:

$$\begin{gathered} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} - \nabla \cdot (\nu(\theta)\mathbf{T}(\mathbf{u})) + \nabla p = \mu_0 (\mathbf{M} \cdot \nabla)\mathbf{H} + \tfrac{\mu_0}{2} \nabla \times (\mathbf{M} \times \mathbf{H}) + \tfrac{\lambda}{\varepsilon}\theta\nabla\psi, \\ \nabla \cdot \mathbf{u} = 0, \end{gathered} \tag{2}$$

where $\mathbf{T}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^\top)$ is the symmetric rate-of-strain tensor, $\nu(\theta) = \nu_+ \frac{1+\theta}{2} + \nu_- \frac{1-\theta}{2} \geq \nu_w > 0$ is the phase-dependent viscosity, $\mu_0 > 0$ is the magnetic permeability of free space, and $\lambda > 0$ is the surface-tension coefficient. The terms on the right-hand side are: the Kelvin body force $\mu_0(\mathbf{M} \cdot \nabla)\mathbf{H}$, the antisymmetric magnetic stress divergence $\frac{\mu_0}{2}\nabla \times (\mathbf{M} \times \mathbf{H})$ (term T3 of [3]), and the Cahn–Hilliard surface-tension force.

Magnetization. The magnetization vector $\mathbf{M} : \Omega \times [0, T_f] \to \mathbb{R}^d$ satisfies the Shliomis relaxation equation [1], including the two terms restored by [3]:

$$\partial_t \mathbf{M} + (\mathbf{u} \cdot \nabla)\mathbf{M} + \tfrac{1}{T}\big(\mathbf{M} - \tilde{\chi}(\theta)\mathbf{H}\big) - \beta\mathbf{M} \times (\mathbf{M} \times \mathbf{H}) + \tfrac{1}{2}(\nabla \times \mathbf{u}) \times \mathbf{M} = \mathbf{0}, \tag{3}$$

where $T > 0$ is the relaxation time, $\tilde{\chi}(\theta) = \chi_0 \frac{1+\theta}{2}$ is the phase-dependent susceptibility with $\chi_0 \in (0, 4)$, and $\beta \geq 0$ is the Landau–Lifshitz damping constant. The term $-\beta\mathbf{M} \times (\mathbf{M} \times \mathbf{H})$ (term T1, Landau–Lifshitz damping) and $\frac{1}{2}(\nabla \times \mathbf{u}) \times \mathbf{M}$ (term T2, spin–vorticity coupling) are the two nonlinear additions relative to the base model of [2].

Magnetostatics. The magnetic scalar potential $\phi : \Omega \times [0, T_f] \to \mathbb{R}$ satisfies:

$$-\Delta\phi = \nabla \cdot (\mathbf{h}_a - \mathbf{M}), \qquad \mathbf{H} := \nabla\phi, \tag{4}$$

with applied field $\mathbf{h}_a : \Omega \times [0, T_f] \to \mathbb{R}^d$ and Neumann boundary condition $\partial_{\mathbf{n}}\phi\big|_\Gamma = (\mathbf{h}_a - \mathbf{M}) \cdot \mathbf{n}\big|_\Gamma$. This gives $\mathbf{H} = \mathbf{h}_a + \nabla\phi_{\text{scatt}}$ with ϕ_{scatt} the scattered potential; we use the convention $\phi =$ total potential, $\mathbf{H} = \nabla\phi$.

Boundary and initial conditions. On Γ we impose:

$$\mathbf{u}\big|_\Gamma = \mathbf{0}, \quad \partial_{\mathbf{n}}\theta\big|_\Gamma = \partial_{\mathbf{n}}\psi\big|_\Gamma = 0, \quad \partial_{\mathbf{n}}\mathbf{M}\big|_\Gamma = \mathbf{0}, \quad \partial_{\mathbf{n}}\phi\big|_\Gamma = (\mathbf{h}_a - \mathbf{M}) \cdot \mathbf{n}. \tag{5}$$

Initial data $\theta_0 \in H^1(\Omega)$, $\mathbf{M}_0 \in [L^2(\Omega)]^d$, $\mathbf{u}_0 \in [L^2(\Omega)]^d$ with $\nabla \cdot \mathbf{u}_0 = 0$ are prescribed.

2.2 Continuous Energy Law

Define the total free energy:

$$E(t) \;:=\; \frac{1}{2}\|\mathbf{u}\|_{L^2}^2 + \frac{\mu_0}{2}\|\mathbf{M}\|_{L^2}^2 + \frac{\mu_0}{2}\|\nabla\phi\|_{L^2}^2 + \frac{\lambda}{2}\|\nabla\theta\|_{L^2}^2 + \frac{\lambda}{\varepsilon^2}\big(F(\theta), 1\big). \tag{6}$$

Smooth solutions of (1)–(4) satisfy the energy identity:

$$\frac{\mathrm{d}}{\mathrm{d}t}E(t) + \mathcal{D}(t) + \frac{2\mu_0\beta}{T^2}\|\mathbf{M} \times \mathbf{H}\|_{L^2}^2 \;=\; \mathcal{F}(t), \tag{7}$$

where the dissipation rate and forcing functional are:

$$\mathcal{D}(t) \;:=\; \|\sqrt{\nu(\theta)}\,\mathbf{T}(\mathbf{u})\|_{L^2}^2 + \frac{\mu_0}{T}\Big(1 - \frac{\chi_0}{4}\Big)\|\mathbf{M}\|_{L^2}^2 + \frac{\mu_0}{2T}\|\nabla\phi\|_{L^2}^2 + \lambda\gamma\|\nabla\psi\|_{L^2}^2, \tag{8}$$

$$\mathcal{F}(t) \;:=\; \frac{\mu_0}{T}\left(\tilde{\chi}(\theta)\mathbf{H}, \mathbf{H}\right). \tag{9}$$

The β-dissipation term $\frac{2\mu_0\beta}{T^2}\|\mathbf{M} \times \mathbf{H}\|^2$ is positive and reflects the additional energy dissipation due to Landau–Lifshitz damping; it vanishes for $\beta = 0$, recovering the base model of [2].

The derivation uses: incompressibility to eliminate advection in NS; the T2–T3 identity $\big(\frac{1}{2}(\nabla \times \mathbf{u}) \times \mathbf{M}, \mathbf{M}\big) = -\big(\mathbf{M} \times \mathbf{H}, \frac{\mu_0}{2}\nabla \times \mathbf{u}\big)$ (antisymmetry); the Poisson equation to convert the $\nabla\phi$-equation to a term involving $\partial_t\mathbf{M}$; and the identity $(\mathbf{M} \times (\mathbf{M} \times \mathbf{H}), \mathbf{H}) = -\|\mathbf{M} \times \mathbf{H}\|^2$, which follows pointwise from $\big(\boldsymbol{a} \times (\boldsymbol{a} \times \boldsymbol{b})\big) \cdot \boldsymbol{b} = -|\boldsymbol{a} \times \boldsymbol{b}|^2$ (Lemma 4.4; cf. [3]).

Remark 2.1 (Positivity of dissipation and the role of $\chi_0 < 4$)**.** The physical dissipation $\mathcal{D}(t)$ is non-negative provided $\chi_0 < 4$, $\nu_w > 0$, and $T, \gamma > 0$, all of which are satisfied for any physical ferrofluid. The condition $\chi_0 < 4$ has a precise mathematical origin: the magnetization relaxation term in the energy identity contributes a coefficient $\frac{\mu_0}{T}(1 - \frac{\chi_0}{4})$ in front of $\|\mathbf{M}\|_{L^2}^2$. This coefficient is positive if and only if $\chi_0 < 4$. For $\chi_0 \geq 4$ the relaxation term would change sign, and the continuous energy law would no longer imply $\|\mathbf{M}\|^2$ is controlled. Physically, $\chi_0 < 4$ corresponds to magnetic susceptibilities below the Langevin saturation threshold; typical laboratory ferrofluids satisfy $\chi_0 \lesssim 1$. The same condition reappears throughout the discrete analysis in Sections 4–5 for the same algebraic reason.

2.3 Continuous Function Spaces and Weak Formulation

Notation. We use standard Sobolev spaces $H^k(\Omega)$, $H_0^1(\Omega)$, $L_0^2(\Omega)$, and their vector-valued counterparts. The $L^2(\Omega)$ inner product is denoted $(\cdot, \cdot)$. The phase field and chemical potential belong to $\theta, \psi \in L^2(0, T_f; H^1(\Omega))$; the velocity to $\mathbf{u} \in L^2(0, T_f; [H_0^1(\Omega)]^d)$; the magnetization to $\mathbf{M} \in L^2(0, T_f; [L^2(\Omega)]^d)$; and the pressure to $p \in L^2(0, T_f; L_0^2(\Omega))$.

Weak formulation. A weak solution of (1)–(4) satisfies, for a.e. $t \in [0, T_f]$ and all admissible test functions:

$$(\partial_t \theta,\, \lambda_1) - (\mathbf{u}\theta,\, \nabla\lambda_1) + \gamma\,(\nabla\psi,\, \nabla\lambda_1) = 0 \qquad \forall \lambda_1 \in H^1(\Omega), \tag{10a}$$

$$(\psi,\, \lambda_2) - \varepsilon\,(\nabla\theta,\, \nabla\lambda_2) + \tfrac{1}{\varepsilon}\left(f'(\theta),\, \lambda_2\right) = 0 \qquad \forall \lambda_2 \in H^1(\Omega), \tag{10b}$$

$$\begin{aligned}&(\partial_t \mathbf{u},\, \mathbf{v}) + ((\mathbf{u}\cdot\nabla)\mathbf{u},\, \mathbf{v}) + (\nu(\theta)\mathbf{T}(\mathbf{u}),\, \mathbf{T}(\mathbf{v})) - (p,\, \nabla\cdot\mathbf{v})\\ &= \mu_0\,((\mathbf{M}\cdot\nabla)\mathbf{H},\, \mathbf{v}) + \tfrac{\mu_0}{2}\,(\mathbf{M}\times\mathbf{H},\, \nabla\times\mathbf{v}) + \tfrac{\lambda}{\varepsilon}\,(\theta\nabla\psi,\, \mathbf{v}) \qquad \forall \mathbf{v} \in [H_0^1(\Omega)]^d,\end{aligned} \tag{10c}$$

$$(\nabla\cdot\mathbf{u},\, q) = 0 \qquad \forall q \in L_0^2(\Omega), \tag{10d}$$

$$\begin{aligned}&(\partial_t \mathbf{M},\, \mathbf{z}) + ((\mathbf{u}\cdot\nabla)\mathbf{M},\, \mathbf{z}) + \tfrac{1}{T}\,(\mathbf{M} - \tilde{\chi}\mathbf{H},\, \mathbf{z})\\ &-\beta\,(\mathbf{M}\times(\mathbf{M}\times\mathbf{H}),\, \mathbf{z}) + \tfrac{1}{2}\,((\nabla\times\mathbf{u})\times\mathbf{M},\, \mathbf{z}) = 0 \qquad \forall \mathbf{z} \in [L^2(\Omega)]^d,\end{aligned} \tag{10e}$$

$$(\nabla\phi,\, \nabla\xi) + (\mathbf{M},\, \nabla\xi) = (\mathbf{h}_a,\, \nabla\xi) \qquad \forall \xi \in H^1(\Omega). \tag{10f}$$

The T3 term is written in the weak form $\frac{\mu_0}{2}\,(\mathbf{M}\times\mathbf{H},\, \nabla\times\mathbf{v})$ (see Remark 3.1 for the discrete analogue).

Remark 2.2 (Regularity and well-posedness)**.** Well-posedness of the continuous system (1)–(4) for $\beta = 0$ was established in [2]. The $\beta > 0$ case follows analogously, using the identity $(\mathbf{M}\times(\mathbf{M}\times\mathbf{H}),\, \mathbf{H}) = -\|\mathbf{M}\times\mathbf{H}\|^2 \le 0$ (test with $\mathbf{H}$, not $\mathbf{M}$; cf. Lemma 4.4), which shows the T1 term contributes additional dissipation in the energy estimate. The present paper does not revisit continuous well-posedness; we refer to [2, 3] for details.

3 Finite Element Discretization

Function Spaces

Let $\Omega \subset \mathbb{R}^d$ $(d = 2, 3)$ be a convex, polyhedral domain with boundary $\Gamma = \partial\Omega$. Fix a quasi-uniform triangulation $\mathcal{T}_h$ of mesh size $h > 0$. We work with the following finite-element spaces, adopting the notation of [2]:

$$\begin{aligned}&\mathcal{G}_h \subset H^1(\Omega), \quad \mathcal{Y}_h \subset H^1(\Omega) && \text{(phase field, chemical potential)},\\ &\mathcal{M}_h \subset [L^2(\Omega)]^d && \text{(magnetization, DG)},\\ &\mathcal{X}_h \subset H^1(\Omega) && \text{(magnetic potential)},\\ &\mathcal{U}_h \subset [H_0^1(\Omega)]^d, \quad \mathcal{P}_h \subset L_0^2(\Omega) && \text{(velocity, pressure)}.\end{aligned} \tag{11}$$

Since $\mathcal{U}_h \subset [H_0^1(\Omega)]^d$, every $\mathbf{V} \in \mathcal{U}_h$ satisfies $\mathbf{V}|_\Gamma = \mathbf{0}$ and in particular $\mathbf{V}\cdot\mathbf{n}|_\Gamma = 0$. This homogeneous normal-trace condition is used in Lemmas 4.1–4.2 and in the integration-by-parts steps throughout the stability proofs. The pair $(\mathcal{U}_h, \mathcal{P}_h)$ satisfies the uniform inf–sup (LBB) condition [4]:

$$\inf_{0\neq Q\in\mathcal{P}_h}\ \sup_{0\neq\mathbf{V}\in\mathcal{U}_h} \frac{(\nabla\cdot\mathbf{V},\, Q)}{\|Q\|_{L^2}\,\|\nabla\mathbf{V}\|_{L^2}} \ge \beta^* > 0,$$

with β^* independent of h. A fundamental structural requirement inherited from [2] is

$$\nabla\mathcal{X}_h \subset \mathcal{M}_h, \tag{12}$$

which holds for the practical choice $\mathcal{X}_h$ = continuous Q_ℓ polynomials and $\mathcal{M}_h$ = discontinuous $Q_{\ell-1}$ polynomials $(\ell \ge 1)$; see [2, Eq. (55)–(56)].

Skew-Symmetric Transport Forms

We use the skew-symmetric DG form for magnetization transport [2, Eq. (57)]:

$$B_h^m(\mathbf{U},\mathbf{V},\mathbf{W}) = \sum_{T\in\mathcal{T}_h}\int_T \left[(\mathbf{U}\cdot\nabla)\mathbf{V}\cdot\mathbf{W}+\tfrac{1}{2}(\nabla\cdot\mathbf{U})\,\mathbf{V}\cdot\mathbf{W}\right]\mathrm{d}x - \sum_{F\in\mathcal{F}^i}\int_F (\mathbf{U}\cdot\mathbf{n}_F)\,[\![\mathbf{V}]\!]\cdot\{\mathbf{W}\}\,\mathrm{d}S, \quad (13)$$

where $\mathcal{F}^i$ denotes the set of interior faces, $[\![\mathbf{V}]\!] = \mathbf{V}^- - \mathbf{V}^+$ is the jump, and $\{\mathbf{W}\} = \frac{1}{2}(\mathbf{W}^- + \mathbf{W}^+)$ is the average. For the Navier–Stokes convection we use the analogous form B_h on $\mathcal{U}_h$ without face terms (continuous velocity space) [2, Eq. (37)].

The Extended Scheme

Let $K > 0$ be the number of time steps, $\tau = T_{\text{final}}/K$, and $t_k = k\tau$. Set $\delta\varphi^k = \varphi^k - \varphi^{k-1}$, $\nu_\Theta = \nu(\Theta^{k-1})$, $\tilde{\chi}_\Theta = \chi(\Theta^{k-1})$. Initialise $\Theta^0, \mathbf{M}^0, \mathbf{U}^0$ by L^2-projection of the continuous initial data [2, Eq. (41)].

The four equations shared by both scheme variants below are, for all $k \in \{1,\dots,K\}$:

$$\left(\delta\Theta^k/\tau,\,\Lambda\right) - \left(\mathbf{U}^{k-1}\Theta^{k-1},\,\nabla\Lambda\right) - \gamma\left(\nabla\Psi^k,\,\nabla\Lambda\right) = 0, \quad (14a)$$

$$\left(\Psi^k,\,\Upsilon\right) + \varepsilon\left(\nabla\Theta^k,\,\nabla\Upsilon\right) + \tfrac{1}{\varepsilon}\left(f'(\Theta^{k-1}),\,\Upsilon\right) + \tfrac{1}{\eta}\left(\delta\Theta^k,\,\Upsilon\right) = 0, \quad (14b)$$

$$\left(\nabla\Phi^k,\,\nabla X\right) + \left(\mathbf{M}^k,\,\nabla X\right) = \left(\mathbf{h}_a^k,\,\nabla X\right), \quad (14c)$$

$$\left(\nabla\cdot\mathbf{U}^k,\,Q\right) = 0, \quad (14d)$$

where $\mathbf{H}^k := \nabla\Phi^k$ and $\mathbf{T}(\mathbf{U}) = \frac{1}{2}(\nabla\mathbf{U} + \nabla\mathbf{U}^\top)$. These equations are identical to [2].

The Scheme: T1, T2, and T3

T2 uses the current Picard-iterate velocity $\mathbf{U}^k$ and magnetization $\mathbf{M}^k$, and T3 is added to the NS equation in its *weak form*:

$$\begin{aligned}\left(\delta\mathbf{M}^k/\tau,\,\mathbf{Z}\right) - B_h^m(\mathbf{U}^k,\mathbf{Z},\mathbf{M}^k) + \tfrac{1}{T}\left(\mathbf{M}^k,\,\mathbf{Z}\right) - \tfrac{1}{T}\left(\tilde{\chi}_\Theta\mathbf{H}^k,\,\mathbf{Z}\right)\\ -\beta\left(\mathbf{M}^{k-1}\times(\mathbf{M}^{k-1}\times\mathbf{H}^k),\,\mathbf{Z}\right) + \tfrac{1}{2}\left((\nabla\times\mathbf{U}^k)\times\mathbf{M}^k,\,\mathbf{Z}\right) = 0,\end{aligned} \quad (15a)$$

$$\begin{aligned}&\left(\delta\mathbf{U}^k/\tau,\,\mathbf{V}\right) + B_h(\mathbf{U}^{k-1},\mathbf{U}^k,\mathbf{V}) + \left(\nu_\Theta\,\mathbf{T}(\mathbf{U}^k),\,\mathbf{T}(\mathbf{V})\right) - \left(P^k,\,\nabla\cdot\mathbf{V}\right)\\ &= \mu_0 B_h^m(\mathbf{V},\mathbf{H}^k,\mathbf{M}^k) + \tfrac{\lambda}{\varepsilon}\left(\Theta^{k-1}\nabla\Psi^k,\,\mathbf{V}\right) + \tfrac{\mu_0}{2}\left(\mathbf{M}^k\times\mathbf{H}^k,\,\nabla\times\mathbf{V}\right),\end{aligned} \quad (15b)$$

together with (14a)–(14d).

Remark 3.1 (Weak form of T3 and DG regularity)**.** Term T3 is written in the NS equation as $\frac{\mu_0}{2}\left(\mathbf{M}^k\times\mathbf{H}^k,\,\nabla\times\mathbf{V}\right)$ rather than the formally equivalent strong form $\frac{\mu_0}{2}\left(\nabla\times(\mathbf{M}^k\times\mathbf{H}^k),\,\mathbf{V}\right)$. The two are equal on the continuous level (integration by parts, using $\mathbf{V}\cdot\mathbf{n}|_\Gamma = 0$), but they differ in the discrete setting: since $\mathbf{M}^k \in \mathcal{M}_h \subset [L^2(\Omega)]^d$ is discontinuous across interior faces, the field $\mathbf{M}^k\times\mathbf{H}^k$ does not belong to $H(\mathrm{curl},\Omega)$, and distributional integration by parts would generate interior face terms involving tangential jumps $[\![\mathbf{n}\times(\mathbf{M}^k\times\mathbf{H}^k)]\!]$. These face terms do not appear in the weak form. By defining T3 directly as $\frac{\mu_0}{2}\left(\mathbf{M}^k\times\mathbf{H}^k,\,\nabla\times\mathbf{V}\right)$, we avoid this regularity issue entirely: this is a standard $L^2(\Omega)$ inner product, well-defined for DG $\mathbf{M}^k$, and requires no regularity of $\mathbf{M}^k$ beyond L^2. This is the correct weak formulation and the form that the assembler should implement.

Remark 3.2 (Velocity time level: transport and T2)**.** The scheme uses $\mathbf{U}^k$ in *three* places in the magnetization equation (15a): the DG transport term $B_h^m(\mathbf{U}^k, \mathbf{Z}, \mathbf{M}^k)$, the T2 coupling $\frac{1}{2}\big((\nabla \times \mathbf{U}^k) \times \mathbf{M}^k, \mathbf{Z}\big)$, and implicitly in the Kelvin-force cross-cancellation in the energy proof (Step 5). Using $\mathbf{U}^k$ in the transport rather than $\mathbf{U}^{k-1}$ is *required* for the Kelvin cancellation: the NS Kelvin force tested with $\mathbf{V} = 2\tau\mathbf{U}^k$ produces $+2\tau\mu_0 B_h^m(\mathbf{U}^k, \mathbf{H}^k, \mathbf{M}^k)$, and the magnetization transport tested with $\mathbf{Z} = 2\tau\mu_0\mathbf{H}^k$ produces $-2\tau\mu_0 B_h^m(\mathbf{U}^k, \mathbf{H}^k, \mathbf{M}^k)$ (using Lemma 4.2 applied to the same $\mathbf{U}^k$). If $\mathbf{U}^{k-1}$ were used in transport instead, the difference $B_h^m(\mathbf{U}^k - \mathbf{U}^{k-1}, \mathbf{H}^k, \mathbf{M}^k) = O(\tau)$ would generate a remainder that breaks unconditionality. All three choices are first-order accurate in time ($\mathbf{U}^k - \mathbf{U}^{k-1} = O(\tau)$), so the formal accuracy is unaffected. Using $\mathbf{U}^k$ in T2 and in the transport is also physically natural: T2 and T3 arise from the antisymmetric part of the magnetic stress tensor [3], and the transport and T2 terms together describe magnetization advection and rotation by the current flow field $\mathbf{U}^k$.

Element Choices and Implementation

Concrete finite element spaces. The abstract spaces in (11) are realised as follows. Let $\mathcal{T}_h$ be a shape-regular, quasi-uniform hexahedral (or tetrahedral) mesh of Ω.

- *Magnetic potential / magnetization.* We use $\mathcal{X}_h = Q_\ell(\mathcal{T}_h)$ (continuous Q_ℓ polynomials, $\ell \geq 1$) and $\mathcal{M}_h = DQ_{\ell-1}(\mathcal{T}_h)$ (discontinuous $Q_{\ell-1}$ polynomials). The inclusion $\nabla\mathcal{X}_h \subset \mathcal{M}_h$ holds by polynomial degree compatibility: if $\xi \in Q_\ell$, then $\nabla\xi \in Q_{\ell-1}^d$, which is contained in $DQ_{\ell-1}$. This is the pair used in [2, Eq. (55)–(56)] and is essential for the energy-stable Kelvin-force cancellation; see Remark 3.3.
- *Velocity / pressure.* We use $\mathcal{U}_h = Q_m(\mathcal{T}_h)^d \cap [H_0^1(\Omega)]^d$ and $\mathcal{P}_h = DQ_{m-1}(\mathcal{T}_h) \cap L_0^2(\Omega)$ for $m \geq 2$ (e.g. Q_2/DQ_1). This pair satisfies the uniform inf–sup (LBB) condition with β^* independent of h [4]. Continuous pressure spaces (Q_1) are also admissible provided the LBB condition holds for the mesh family.
- *Phase field / chemical potential.* We use $\mathcal{G}_h = \mathcal{Y}_h = Q_r(\mathcal{T}_h)$ for $r \geq 1$. Both fields share the same space, consistent with the H^1-H^{-1} variational structure of the Cahn–Hilliard system.

In the numerical experiments of this paper we take $\ell = m = r = 1$ (Q_1/DQ_0 for magnetics, Q_2/DQ_1 for flow, Q_1/Q_1 for CH) unless otherwise stated.

Remark 3.3 (Necessity of the inclusion $\nabla\mathcal{X}_h \subset \mathcal{M}_h$)**.** The inclusion (12) is not a mere convenience; it is required for the discrete Kelvin-force cancellation in the energy proof (Lemma 4.2). Without it, the term $\mu_0 B_h^m(\mathbf{V}, \mathbf{H}^k, \mathbf{M}^k)$ tested with $\mathbf{V} = 2\tau\mathbf{U}^k$ and the term from the magnetization equation tested with $\mathbf{Z} = 2\tau\mu_0\mathbf{H}^k = 2\tau\mu_0\nabla\Phi^k$ would not cancel exactly, since $\nabla\Phi^k \notin \mathcal{M}_h$ would not be a valid DG test function. Any implementation that chooses $\mathcal{X}_h$ and $\mathcal{M}_h$ independently, without verifying this inclusion, will obtain a non-energy-stable scheme.

Mesh regularity. We assume $\mathcal{T}_h$ is shape-regular (minimum angle condition) and quasi-uniform (mesh size $h \sim h_T$ for all elements T). Shape-regularity ensures validity of the inverse inequality

$$\|\nabla v_h\|_{L^\infty(T)} \;\leq\; C_{\text{inv}}\, h^{-d/2}\, \|v_h\|_{L^2(T)} \quad \forall v_h \in \mathcal{M}_h,\ \forall T \in \mathcal{T}_h, \tag{16}$$

with C_{inv} depending only on the shape-regularity constant and polynomial degree, not on h. This estimate is used in the existence proof (Section 5) to control the T2 nonlinearity. Quasi-uniformity is used in the convergence analysis (Section 6) for the DG face-term estimates.

Solver strategy. At each time step k the scheme is solved as follows:

1. *Cahn–Hilliard step*: solve the coupled (Θ^k, Ψ^k) system (14a)–(14b) (linear, since $f'(\Theta^{k-1})$ is lagged).

2. *Coupled magnetic–flow step*: solve the coupled $(\mathbf{M}^k, \Phi^k, \mathbf{U}^k, P^k)$ system. The Poisson equation is linear in $\mathbf{M}^k$ and can be eliminated analytically (Section 5.1), reducing the nonlinear system to $(\mathbf{U}^k, \mathbf{M}^k)$. A fixed-point (Picard) iteration on the residual converges under the timestep restriction $\tau \le \tau_*(h, R)$ of Theorem 5.4.

Remark 3.4 (Linear solvers)**.** The CH and Poisson subsystems are symmetric positive definite; the linearized magnetization step (within the Picard iteration) is also symmetric positive definite. The Navier–Stokes step is a saddle-point problem. For moderate problem sizes, direct factorization (e.g. UMFPACK, MUMPS) is used. For large-scale parallel computations, block-preconditioned iterative methods (AMG for scalar blocks, block-triangular preconditioner for NS) are more practical; the energy-stable structure of the scheme ensures well-conditioned subsystems.

Implementation of B_h^m. The DG transport form (13) is assembled by looping over all cells $T \in \mathcal{T}_h$ for the volume terms and over all interior faces $F \in \mathcal{F}^i$ for the face terms. For the face terms, the upstream velocity $(\mathbf{U} \cdot \mathbf{n}_F)^- = \min(\mathbf{U} \cdot \mathbf{n}_F, 0)$ is used in the upwind-biased formulation (13), consistent with [2]. The form is skew-symmetric: $B_h^m(\mathbf{U}, \mathbf{V}, \mathbf{W}) = -B_h^m(\mathbf{U}, \mathbf{W}, \mathbf{V})$ when $\mathbf{U} \cdot \mathbf{n}|_\Gamma = 0$ (Lemma 4.2), which is exploited in the energy proofs.

4 Discrete Energy Stability

4.1 Notation, Cross Products in 2D, and Energy Functionals

All analysis is carried out for $d \in \{2, 3\}$. In $d = 3$, the cross product $\mathbf{a} \times \mathbf{b}$ is the standard $\mathbb{R}^3$ operation. In $d = 2$, we adopt the embedding

$$\mathbf{a} = (a_1, a_2) \;\mapsto\; \hat{\mathbf{a}} = (a_1, a_2, 0) \in \mathbb{R}^3, \tag{17}$$

and define $\mathbf{a} \times \mathbf{b} := \hat{\mathbf{a}} \times \hat{\mathbf{b}} = (0, 0, a_1 b_2 - a_2 b_1)$, so that $|\mathbf{a} \times \mathbf{b}|^2 = (a_1 b_2 - a_2 b_1)^2$. The scalar curl is $\nabla \times \mathbf{a} = \partial_1 a_2 - \partial_2 a_1$, and the term $(\nabla \times \mathbf{a}) \times \mathbf{b}$ is interpreted as $(0, 0, \nabla \times \mathbf{a}) \times (b_1, b_2, 0)$. All vector identities used below hold under this convention.

The discrete energy at time level k is

$$\mathcal{E}^k := \frac{1}{2}\left\|\mathbf{U}^k\right\|^2_{L^2(\Omega)} + \frac{\mu_0}{2}\left\|\mathbf{M}^k\right\|^2_{L^2(\Omega)} + \frac{\mu_0}{2}\left\|\nabla\Phi^k\right\|^2_{L^2(\Omega)} + \frac{\lambda}{2}\left\|\nabla\Theta^k\right\|^2_{L^2(\Omega)} + \frac{\lambda}{\varepsilon^2}\big(F(\Theta^k), 1\big). \tag{18}$$

The numerical dissipation increment and physical dissipation rate are

$$\mathcal{D}_n^k := \frac{1}{2}\left\|\delta\mathbf{U}^k\right\|^2_{L^2(\Omega)} + \frac{\mu_0}{2}\left\|\delta\mathbf{M}^k\right\|^2_{L^2(\Omega)} + \frac{\mu_0}{2}\left\|\delta\nabla\Phi^k\right\|^2_{L^2(\Omega)} + \frac{\lambda}{2}\left\|\delta\nabla\Theta^k\right\|^2_{L^2(\Omega)}, \tag{19}$$

$$\mathcal{D}_p^k := \frac{\mu_0}{T}\Big(1 - \frac{\chi_0}{4}\Big)\left\|\mathbf{M}^k\right\|^2_{L^2(\Omega)} + \frac{\mu_0}{2T}\left\|\nabla\Phi^k\right\|^2_{L^2(\Omega)} + \left\|\sqrt{\nu_\Theta}\,\mathbf{T}(\mathbf{U}^k)\right\|^2_{L^2(\Omega)} + \frac{\lambda\gamma}{\varepsilon}\left\|\nabla\Psi^k\right\|^2_{L^2(\Omega)}, \tag{20}$$

and the external forcing functional is

$$\mathcal{F}^k := \frac{\mu_0}{T}\left\|\mathbf{h}_a^k\right\|^2_{L^2(\Omega)} + \frac{\mu_0 T}{\tau}\int_{t_{k-1}}^{t_k} \|\partial_t \mathbf{h}_a\|^2_{L^2(\Omega)}\,\mathrm{d}s. \tag{21}$$

The condition $\chi_0 \le 4$ ensures $\mathcal{D}_p^k > 0$; here $\chi_0 = \max_{\Theta\in[-1,1]} \chi(\Theta)$ is the material susceptibility constant.

4.2 Preliminary Lemmas

Lemma 4.1 (Skew-symmetry of B_h^m, [2, Eq. (57)])**.** *Let* $\mathbf{U} \in \mathcal{U}_h$ *with* $\mathbf{U} \cdot \mathbf{n}|_\Gamma = 0$. *Then* $B_h^m(\mathbf{U}, \mathbf{V}, \mathbf{V}) = 0$ *for all* $\mathbf{V} \in \mathcal{M}_h$.

Proof. Set $\mathbf{W} = \mathbf{V}$ in (13). Since $(\mathbf{U} \cdot \nabla)\mathbf{V} \cdot \mathbf{V} = \frac{1}{2}\mathbf{U} \cdot \nabla|\mathbf{V}|^2$, integration by parts cell-wise gives $\int_T (\mathbf{U} \cdot \nabla)\mathbf{V} \cdot \mathbf{V}\, \mathrm{d}x = \frac{1}{2}\int_{\partial T}(\mathbf{U} \cdot \mathbf{n})|\mathbf{V}|^2\, \mathrm{d}S - \frac{1}{2}\int_T (\nabla \cdot \mathbf{U})|\mathbf{V}|^2\, \mathrm{d}x$. The cell-interior terms cancel. On each interior face, the identity $[\![\mathbf{V}]\!] \cdot \{\mathbf{V}\} = \frac{1}{2}(|\mathbf{V}^-|^2 - |\mathbf{V}^+|^2)$ shows that the face sum equals the boundary term from the cell integrals, and these cancel globally. The outer boundary term vanishes since $\mathbf{U} \cdot \mathbf{n}|_\Gamma = 0$. □

Lemma 4.2 (Skew-antisymmetry of B_h^m)**.** *Let* $\mathbf{U} \in \mathcal{U}_h$ *with* $\mathbf{U} \cdot \mathbf{n}|_\Gamma = 0$. *Then for all* $\mathbf{V}, \mathbf{W} \in \mathcal{M}_h$*:*

$$B_h^m(\mathbf{U}, \mathbf{V}, \mathbf{W}) + B_h^m(\mathbf{U}, \mathbf{W}, \mathbf{V}) = 0. \tag{22}$$

This holds without *any assumption on* $\nabla \cdot \mathbf{U}$ *and* without *any continuity of* $\mathbf{V}$ *or* $\mathbf{W}$ *across faces.*

Proof. Add $B_h^m(\mathbf{U}, \mathbf{V}, \mathbf{W})$ and $B_h^m(\mathbf{U}, \mathbf{W}, \mathbf{V})$ via (13). **Volume terms:**

$$(\mathbf{U} \cdot \nabla)\mathbf{V} \cdot \mathbf{W} + \tfrac{1}{2}(\nabla \cdot \mathbf{U})(\mathbf{V} \cdot \mathbf{W}) + (\mathbf{U} \cdot \nabla)\mathbf{W} \cdot \mathbf{V} + \tfrac{1}{2}(\nabla \cdot \mathbf{U})(\mathbf{W} \cdot \mathbf{V}) = \nabla \cdot \big(\mathbf{U}(\mathbf{V} \cdot \mathbf{W})\big),$$

since $(\mathbf{U} \cdot \nabla)\mathbf{V} \cdot \mathbf{W} + (\mathbf{U} \cdot \nabla)\mathbf{W} \cdot \mathbf{V} = \mathbf{U} \cdot \nabla(\mathbf{V} \cdot \mathbf{W})$. Summing over cells and applying the divergence theorem: $\sum_T \int_T \nabla \cdot (\mathbf{U}(\mathbf{V} \cdot \mathbf{W}))\, \mathrm{d}x = \sum_T \int_{\partial T}(\mathbf{U} \cdot \mathbf{n})(\mathbf{V} \cdot \mathbf{W})\, \mathrm{d}S$. **Face terms:** Using the DG product rule $[\![\mathbf{V} \cdot \mathbf{W}]\!] = [\![\mathbf{V}]\!] \cdot \{\mathbf{W}\} + \{\mathbf{V}\} \cdot [\![\mathbf{W}]\!]$, the face sum equals $\sum_F \int_F (\mathbf{U} \cdot \mathbf{n}_F)[\![\mathbf{V} \cdot \mathbf{W}]\!]\, \mathrm{d}S$. **Cancellation:** Each interior face appears in the cell boundary sum twice with opposite orientations, so the cell sum $\sum_T \int_{\partial T}$ and the face sum cancel exactly. The outer boundary term is zero because $\mathbf{U} \cdot \mathbf{n}|_\Gamma = 0$. □

Remark 4.3 (On the continuity of $\nabla\Phi$ for C^0 potentials)**.** A natural but incorrect argument would claim $[\![\mathbf{H}]\!] = 0$ on interior faces from the fact that $\Phi \in \mathcal{X}_h \subset C^0(\Omega)$. This is false: C^0 continuity of Φ implies only that Φ is single-valued at nodes and edges, not that $\nabla\Phi$ is continuous across faces. For standard Q_ℓ elements, $\nabla\Phi$ is piecewise $Q_{\ell-1}$ and its components are genuinely discontinuous at interior faces; in general $[\![\nabla\Phi]\!] = O(h)$ and is not zero pointwise. Lemma 4.2 avoids this issue entirely: the Kelvin-force cancellation follows from the algebraic antisymmetry of B_h^m, which holds for any $\mathbf{V}, \mathbf{W} \in \mathcal{M}_h$ with no regularity requirement on $\mathbf{H}$.

Lemma 4.4 (Pointwise vector identities)**.** *For any* $\mathbf{a}, \mathbf{b}, \mathbf{c}$ *in* $\mathbb{R}^3$*, or in* $\mathbb{R}^2$ *under embedding* (17)*:*

$$\big[\mathbf{a} \times (\mathbf{a} \times \mathbf{b})\big] \cdot \mathbf{b} = -|\mathbf{a} \times \mathbf{b}|^2 \;\le\; 0, \tag{23}$$

$$\big[(\mathbf{c} \times \mathbf{a})\big] \cdot \mathbf{a} = 0. \tag{24}$$

Proof. (23): By BAC–CAB, $\mathbf{a} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\mathbf{a} \cdot \mathbf{b}) - \mathbf{b}|\mathbf{a}|^2$, so $[\mathbf{a} \times (\mathbf{a} \times \mathbf{b})] \cdot \mathbf{b} = (\mathbf{a} \cdot \mathbf{b})^2 - |\mathbf{a}|^2|\mathbf{b}|^2 = -|\mathbf{a} \times \mathbf{b}|^2$. (24): By the scalar triple product, $(\mathbf{c} \times \mathbf{a}) \cdot \mathbf{a} = \det[\mathbf{c}, \mathbf{a}, \mathbf{a}] = 0$. □

4.3 Unconditional Energy Stability

We now state and prove the main stability theorem. The hypothesis $\beta T \le 1$ below is a physical constraint of the Shliomis model: $\beta = 1/(6\eta_0 V_p)$ with η_0 the fluid viscosity and V_p the magnetic-particle volume, while T is the Néel–Brown relaxation time; βT measures the ratio of rotational drag to relaxation and is well below unity for physically relevant ferrofluids [1]. The stability *itself* is unconditional in h and τ.

Theorem 4.5 (Unconditional energy stability)**.** *Let the scheme* (15) *(together with the shared equations* (14a)–(14d)*) hold. Assume*

$$\nabla \mathcal{X}_h \subset \mathcal{M}_h, \qquad \eta \le \varepsilon, \qquad \chi_0 \le 4, \qquad \nu_\Theta \ge \nu_w > 0, \qquad \beta \ge 0, \qquad \beta T \ \le \ 1. \tag{25}$$

Then for every $k \ge 1$:

$$\boxed{\mathcal{E}^k + \mathcal{D}_n^k + \tau\, \mathcal{D}_p^k + 2\mu_0 \beta \tau \left\| \mathbf{M}^{k-1} \times \mathbf{H}^k \right\|_{L^2(\Omega)}^2 \ \le \ \mathcal{E}^{k-1} + \tau\, \mathcal{F}^k.} \tag{26}$$

There is no remainder term. Summing over $k = 1, \dots, K$:

$$\mathcal{E}^K + \sum_{k=1}^{K} (\mathcal{D}_n^k + \tau\, \mathcal{D}_p^k) + 2\mu_0 \beta \tau \sum_{k=1}^{K} \left\| \mathbf{M}^{k-1} \times \mathbf{H}^k \right\|_{L^2(\Omega)}^2 \ \le \ \mathcal{E}^0 + \tau \sum_{k=1}^{K} \mathcal{F}^k. \tag{27}$$

The estimate is unconditional*: it holds for all $\tau > 0$ and all $h > 0$ without any regularity assumption on $\mathbf{U}^{k-1}$. The β-term strengthens the energy law relative to the Nochetto base scheme [2, Prop. 4.1].*

Proof of Theorem 4.5. Fix $k \ge 1$ and choose simultaneous test functions:

$$\Lambda = \tfrac{2\lambda\tau}{\varepsilon} \Psi^k, \quad \Upsilon = \tfrac{2\lambda}{\varepsilon} \delta\Theta^k, \quad \mathbf{V} = 2\tau \mathbf{U}^k, \quad \mathbf{Z} = 2\tau\mu_0 \mathbf{M}^k, \quad X = 2\tau\mu_0 \Phi^k. \tag{28}$$

Step 1: Cahn–Hilliard. Equations (14a)–(14b) are unchanged from [2]. Using the polarization identity $2(a, a-b) = |a|^2 - |b|^2 + |a-b|^2$ and the Eyre splitting of F:

$$\lambda \left\| \nabla\Theta^k \right\|_{L^2(\Omega)}^2 - \lambda \left\| \nabla\Theta^{k-1} \right\|_{L^2(\Omega)}^2 + \lambda \left\| \delta\nabla\Theta^k \right\|_{L^2(\Omega)}^2 + \tfrac{2\lambda}{\varepsilon^2} (F(\Theta^k) - F(\Theta^{k-1}), 1) + \tfrac{2\lambda\gamma\tau}{\varepsilon} \left\| \nabla\Psi^k \right\|_{L^2(\Omega)}^2 \ \le \ \mathcal{S}_{\text{cap}}, \tag{29}$$

where $\mathcal{S}_{\text{cap}} = \frac{2\lambda\tau}{\varepsilon} \left(\mathbf{U}^{k-1}\Theta^{k-1}, \nabla\Psi^k \right)$ will cancel the capillary force in the NS step.

Step 2: Navier–Stokes. Substituting $\mathbf{V} = 2\tau\mathbf{U}^k$ in (15b), the polarization identity and $B_h(\mathbf{U}^{k-1}, \mathbf{U}^k, \mathbf{U}^k) = 0$ give:

$$\left\| \mathbf{U}^k \right\|_{L^2(\Omega)}^2 - \left\| \mathbf{U}^{k-1} \right\|_{L^2(\Omega)}^2 + \left\| \delta\mathbf{U}^k \right\|_{L^2(\Omega)}^2 + 2\tau \left\| \sqrt{\nu_\Theta}\, \mathbf{T}(\mathbf{U}^k) \right\|_{L^2(\Omega)}^2 = 2\tau\mu_0 B_h^m(\mathbf{U}^k, \mathbf{H}^k, \mathbf{M}^k) + \mathcal{S}'_{\text{cap}} + \mathcal{T}_3, \tag{30}$$

where $\mathcal{S}'_{\text{cap}} = -\frac{2\lambda\tau}{\varepsilon} \left(\Theta^{k-1}\nabla\Psi^k, \mathbf{U}^k \right)$ cancels with $\mathcal{S}_{\text{cap}}$ as in [2], and $\mathcal{T}_3 = \tau\mu_0 \left(\mathbf{M}^k \times \mathbf{H}^k, \nabla \times \mathbf{U}^k \right)$ is the T3 contribution; cf. Step 2' below.

Step 3a: Magnetization, test $\mathbf{Z} = 2\tau\mu_0\mathbf{M}^k$. From (15a), using $B_h^m(\mathbf{U}^k, \mathbf{M}^k, \mathbf{M}^k) = 0$ (Lemma 4.1):

$$\mu_0 \Big(\left\| \mathbf{M}^k \right\|_{L^2(\Omega)}^2 - \left\| \mathbf{M}^{k-1} \right\|_{L^2(\Omega)}^2 + \left\| \delta\mathbf{M}^k \right\|_{L^2(\Omega)}^2 \Big) + \tfrac{2\mu_0\tau}{T} \left\| \mathbf{M}^k \right\|_{L^2(\Omega)}^2 = \tfrac{2\mu_0\tau}{T} \left(\tilde{\chi}_\Theta \mathbf{H}^k, \mathbf{M}^k \right) + \mathrm{T1}|_{\mathbf{M}^k} + \mathrm{T2}|_{\mathbf{M}^k}. \tag{31}$$

T1 with $\mathbf{M}^k$: By Cauchy–Schwarz and Young's inequality with $\mathbf{a} = \mathbf{M}^{k-1}$, $\mathbf{b} = \mathbf{H}^k$:

$$2\mu_0\tau\beta \left| \left(\mathbf{M}^{k-1} \times (\mathbf{M}^{k-1} \times \mathbf{H}^k), \mathbf{M}^k \right) \right| \ \le \ \mu_0\tau\beta \left\| \mathbf{M}^{k-1} \times \mathbf{H}^k \right\|_{L^2(\Omega)}^2 + \mu_0\tau\beta \left\| \mathbf{M}^k \right\|_{L^2(\Omega)}^2. \tag{32}$$

The $\left\| \mathbf{M}^k \right\|_{L^2(\Omega)}^2$ term on the right is absorbed by the $\frac{2\mu_0\tau}{T} \left\| \mathbf{M}^k \right\|_{L^2(\Omega)}^2$ term on the left under the hypothesis $\beta T \le 1$ (consuming half of it). This is a Young inequality bound, not a sign-definite dissipation; the genuine β-dissipation $+2\mu_0\tau\beta \left\| \mathbf{M}^{k-1} \times \mathbf{H}^k \right\|_{L^2(\Omega)}^2$ comes from the $\mathbf{H}^k$-test in Step 3b below.

Step 3a': T2 self-cancellation with $\mathbf{M}^k$.

With T2 using $\mathbf{U}^k$ and $\mathbf{M}^k$ from (15a), testing with $\mathbf{Z} = 2\tau\mu_0\mathbf{M}^k$:

$$-\mu_0\tau\left((\nabla\times\mathbf{U}^k)\times\mathbf{M}^k,\,\mathbf{M}^k\right) \;=\; 0, \tag{33}$$

by Lemma 4.4(24) with $\mathbf{c} = \nabla\times\mathbf{U}^k$, $\mathbf{a} = \mathbf{M}^k$, applied *pointwise* at each quadrature point.

Step 3b: Magnetization, test $\mathbf{Z} = 2\tau\mu_0\mathbf{H}^k$.

This is valid since $\mathbf{H}^k = \nabla\Phi^k \in \nabla\mathcal{X}_h \subset \mathcal{M}_h$ (hypothesis (25), inclusion (12)).
T1 with $\mathbf{H}^k$: By Lemma 4.4(23) with $\mathbf{a} = \mathbf{M}^{k-1}$, $\mathbf{b} = \mathbf{H}^k$:

$$-2\mu_0\tau\beta\left(\mathbf{M}^{k-1}\times(\mathbf{M}^{k-1}\times\mathbf{H}^k),\,\mathbf{H}^k\right) \;=\; +2\mu_0\tau\beta\left\|\mathbf{M}^{k-1}\times\mathbf{H}^k\right\|^2_{L^2(\Omega)}. \tag{34}$$

This is the key dissipative contribution from T1: it appears on the left-hand side of (26) with a positive sign.

Step 2': T3 in the NS equation.

T3 is defined in weak form in (15b). Testing with $\mathbf{V} = 2\tau\mathbf{U}^k$ gives directly, by definition:

$$+\tau\mu_0\left(\mathbf{M}^k\times\mathbf{H}^k,\,\nabla\times\mathbf{U}^k\right). \tag{35}$$

No integration by parts is needed and no regularity of $\mathbf{M}^k\times\mathbf{H}^k$ beyond $L^2(\Omega)$ is required; see Remark 3.1. In $d = 2$, this reads $\tau\mu_0\int_\Omega(M_1^kH_2^k - M_2^kH_1^k)(\partial_1U_2^k - \partial_2U_1^k)\,\mathrm{d}x$.

Step 3b': T2 with $\mathbf{H}^k$ test.

Testing T2 (15a) with $\mathbf{Z} = 2\tau\mu_0\mathbf{H}^k$:

$$-\mu_0\tau\left((\nabla\times\mathbf{U}^k)\times\mathbf{M}^k,\,\mathbf{H}^k\right). \tag{36}$$

By the scalar triple product identity $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c} = \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$:

$$\left((\nabla\times\mathbf{U}^k)\times\mathbf{M}^k,\,\mathbf{H}^k\right) = \left(\nabla\times\mathbf{U}^k,\,\mathbf{M}^k\times\mathbf{H}^k\right). \tag{37}$$

Cross-cancellation of T2 and T3.

Adding (35) (T3 from NS, Step 2') and (37) (T2 from Mag, Step 3b'):

$$+\,\tau\mu_0\underbrace{\left(\mathbf{M}^k\times\mathbf{H}^k,\,\nabla\times\mathbf{U}^k\right)}_{\text{T3 from NS (35)}} - \tau\mu_0\underbrace{\left(\nabla\times\mathbf{U}^k,\,\mathbf{M}^k\times\mathbf{H}^k\right)}_{\text{T2 from Mag (37)}} \;=\; 0. \tag{38}$$

The cancellation is *exact and purely algebraic*: the two inner products are equal by symmetry of the $L^2(\Omega)$ pairing. No integration by parts is involved, no assumption on $[\![\mathbf{H}^k]\!]$ is needed, no divergence-free condition on $\mathbf{U}^k$ is required, and no $H(\mathrm{curl})$ regularity of $\mathbf{M}^k\times\mathbf{H}^k$ is assumed. The use of the weak form for T3 in (15b) is precisely what makes this argument valid for DG $\mathbf{M}^k$.

Step 5': Kelvin force cancellation (exact).

In Scheme II, the magnetization transport uses $\mathbf{U}^k$ (see (15a) and Remark 3.2). From Step 2, the NS Kelvin force tested with $\mathbf{V} = 2\tau\mathbf{U}^k$ contributes $+2\tau\mu_0B_h^m(\mathbf{U}^k,\mathbf{H}^k,\mathbf{M}^k)$. The magnetization transport tested with $\mathbf{Z} = 2\tau\mu_0\mathbf{H}^k$ (using $-B_h^m(\mathbf{U}^k,\mathbf{Z},\mathbf{M}^k)$ from (15a)) contributes $-2\tau\mu_0B_h^m(\mathbf{U}^k,\mathbf{H}^k,\mathbf{M}^k)$. The two contributions share the same argument order $(\mathbf{U}^k,\mathbf{H}^k,\mathbf{M}^k)$ and equal magnitude with opposite signs:

$$+2\tau\mu_0B_h^m(\mathbf{U}^k,\mathbf{H}^k,\mathbf{M}^k) \;-\; 2\tau\mu_0B_h^m(\mathbf{U}^k,\mathbf{H}^k,\mathbf{M}^k) \;=\; 0, \tag{39}$$

so they cancel exactly without any further identity. No transport mismatch of the form $B_h^m(\mathbf{U}^k - \mathbf{U}^{k-1}, \mathbf{H}^k, \mathbf{M}^k) = O(\tau)$ arises, because both NS and the magnetization transport are tested at the same velocity time level $\mathbf{U}^k$. This is the principal reason the scheme is unconditionally stable.

(Skew-antisymmetry of B_h^m in the second and third arguments (Lemma 4.2) is still used elsewhere — in particular to derive the cell-and-face cancellations behind $B_h^m(\mathbf{U}, \mathbf{V}, \mathbf{V}) = 0$ (Lemma 4.1) — but is not needed here.)

Step 4: Poisson equation and summation by parts. Choosing $X = 2\tau\mu_0\Phi^k$ in (14c) and applying the summation-by-parts formula [2, Eq. (75)] handles the cross-term $\sum_k \left(\delta\mathbf{M}^k, \nabla\Phi^k\right)$ and produces the $\left\|\nabla\Phi^k\right\|_{L^2(\Omega)}^2$ terms in $\mathcal{E}^k$ and $\mathcal{D}_n^k$; see [2, pp. 510–511].

Assembly. Combining Steps 1–4: Step 1 contributes the CH terms, Step 2 contributes the kinetic and viscous terms (with the T3 source on the right), Step 3a/3a' contribute the magnetization mass and relaxation terms (with T2 self-cancellation), Step 3b contributes the magnetostatic terms and the genuine β-dissipation $+2\mu_0\tau\beta\left\|\mathbf{M}^{k-1}\times\mathbf{H}^k\right\|_{L^2(\Omega)}^2$, and Step 4 produces $\left\|\nabla\Phi^k\right\|_{L^2(\Omega)}^2$. The capillary terms $\mathcal{S}_{\text{cap}}, \mathcal{S}'_{\text{cap}}$ cancel between Steps 1 and 2. The Kelvin mismatch vanishes (Step 5'), and the T2–T3 cross-term cancels (Eq. (38)). The $\left\|\mathbf{M}^k\right\|_{L^2(\Omega)}^2$ remainder from Step 3a (T1 with $\mathbf{M}^k$ test) is absorbed under $\beta T \le 1$. Assembling all contributions yields (26) with no remainder. The β-dissipation (34) stands on the left-hand side, providing the additional unconditional stability $+2\mu_0\beta\tau\|\mathbf{M}^{k-1}\times\mathbf{H}^k\|^2$ beyond the Nochetto base scheme. □

Remark 4.6 (Comparison with Zhang–He–Yang)**.** Zhang et al. [3] achieve the same T2–T3 cross-cancellation using auxiliary variables $\tilde{\boldsymbol{m}}^n$, $\tilde{\mathbf{H}}^n$ in a two-step scheme for a continuous Galerkin magnetization space. Our proof uses no auxiliary variables; the cancellation follows directly from the L^2 symmetry of the inner product and the scalar triple product identity. Both approaches require T2 and T3 to be evaluated at the same velocity time level.

Corollary 4.7 (Mass conservation)**.** *The scheme preserves phase-field mass:* $(\Theta^k, 1) = (\Theta^0, 1)$ *for all* k.

Proof. Set $\Lambda = 1$ in (14a); all other terms vanish. Terms T1–T3 do not appear in the CH equations. □

5 Existence and Uniqueness

At each time step k, the data from the previous step, $\mathcal{D}^{k-1} := (\Theta^{k-1}, \Psi^{k-1}, \mathbf{M}^{k-1}, \Phi^{k-1}, \mathbf{U}^{k-1}, P^{k-1})$, is assumed given in the respective discrete spaces. We prove that each scheme determines a solution $(\Theta^k, \Psi^k, \mathbf{M}^k, \Phi^k, \mathbf{U}^k, P^k) \in \mathcal{G}_h \times \mathcal{Y}_h \times \mathcal{M}_h \times \mathcal{X}_h \times \mathcal{U}_h \times \mathcal{P}_h$ at each step k, and determine the conditions under which the solution is unique.

5.1 Reduction via the Poisson Subproblem

Both schemes share the magnetostatic equation (14c):

$$\left(\nabla\Phi^k, \nabla X\right) = \left(\mathbf{h}_a^k - \mathbf{M}^k, \nabla X\right) \quad \forall\, X \in \mathcal{X}_h.$$

For any fixed $\mathbf{M}^k \in \mathcal{M}_h$, this is a coercive symmetric bilinear form on $\mathcal{X}_h$ (after fixing the mean of Φ^k to zero). By the Lax–Milgram theorem in the finite-element space $\mathcal{X}_h$, this problem has a unique solution $\Phi^k = \Phi^k(\mathbf{M}^k)$, and the map $\mathbf{M}^k \mapsto \Phi^k(\mathbf{M}^k)$ is linear and continuous. Setting $\mathbf{H}^k = \nabla\Phi^k(\mathbf{M}^k)$, we may treat $\mathbf{H}^k$ as a known function of $\mathbf{M}^k$ throughout.

Similarly, the Cahn–Hilliard equations (14a)–(14b) are linear in (Θ^k, Ψ^k) for known $\mathbf{U}^{k-1}$, Θ^{k-1}, and have a unique solution by the same argument.

The fixed-point analysis below therefore operates on the reduced space $\mathcal{U}_h \times \mathcal{M}_h$ only.

5.2 The Linearized Step (Picard Iterate)

The genuinely nonlinear obstacle to existence is the T2 term $\frac{1}{2}\big((\nabla\times\mathbf{U}^k)\times\mathbf{M}^k,\,\mathbf{Z}\big)$ in (15a), which is bilinear in the current unknowns $(\mathbf{U}^k,\mathbf{M}^k)$. All other terms in the system are linear in the current unknowns (with coefficients evaluated at the previous step). Given a frozen pair $(\tilde{\mathbf{U}},\tilde{\mathbf{M}})$, the *linearized* step at time t_k replaces T2 by the known source $\frac{1}{2}\Big((\nabla\times\tilde{\mathbf{U}})\times\tilde{\mathbf{M}},\,\mathbf{Z}\Big)$ and asks for $(\mathbf{U}^k,\mathbf{M}^k,\Phi^k,P^k)$ solving the resulting linear system together with the CH equations (14a)–(14b) for (Θ^k,Ψ^k).

Lemma 5.1 (Well-posedness of the linearized step)**.** *Under hypothesis* (25)*, for any fixed pair* $(\tilde{\mathbf{U}},\tilde{\mathbf{M}})\in\mathcal{U}_h\times\mathcal{M}_h$ *the linearized step has a unique solution* $(\Theta^k,\Psi^k,\mathbf{M}^k,\Phi^k,\mathbf{U}^k,P^k)$.

Proof. It suffices to show that the homogeneous system (zero data $\mathcal{D}^{k-1}=\mathbf{0}$, $\mathbf{h}_a^k=\mathbf{0}$, and zero T2 source) has only the zero solution; existence then follows from the rank–nullity theorem since the system is square and linear. Apply the energy estimate of Theorem 4.5 with $\mathcal{E}^0=0$, $\mathcal{F}^k=0$ (the T2 *source* plays no role in the proof of energy stability for the linearized step beyond what its absorption with the relaxation term already gives, and vanishes for zero data):

$$\tfrac{1}{2}\left\|\mathbf{U}^k\right\|^2_{L^2(\Omega)}+\tfrac{\mu_0}{2}\left\|\mathbf{M}^k\right\|^2_{L^2(\Omega)}+\tfrac{\mu_0}{2}\left\|\nabla\Phi^k\right\|^2_{L^2(\Omega)}+\mathcal{D}_n^k+\tau\,\mathcal{D}_p^k\;\le\;0. \tag{40}$$

Every term is non-negative, hence zero. So $\mathbf{U}^k=\mathbf{0}$, $\mathbf{M}^k=\mathbf{0}$, $\Phi^k=0$ (mean-zero normalization), $\Theta^k=0$. The pressure $P^k=0$ follows from the LBB condition for $(\mathcal{U}_h,\mathcal{P}_h)$ applied to the reduced NS equation $-\big(P^k,\,\nabla\cdot\mathbf{V}\big)=0$. Finally $\Psi^k=0$ follows from (14b) with $\Theta^k=0$. □

5.3 Existence via Brouwer Fixed Point

T2 introduces a genuinely nonlinear coupling at the current time level. Existence is established via the Brouwer fixed-point theorem. Throughout this subsection we write

$$\mathcal{Z}_h\;:=\;\mathcal{U}_h\times\mathcal{M}_h \tag{41}$$

for the product space (note: distinct from the chemical-potential space $\mathcal{Y}_h$ in (11)), and we note that all constants C, $\widetilde{C}$, C_0 depend only on physical parameters (ν_w, μ_0, T, χ_0) and mesh shape-regularity, but are independent of h and τ.

The fixed-point map

Given $(\tilde{\mathbf{U}},\tilde{\mathbf{M}})\in\mathcal{Z}_h$, let $(\mathbf{U}^k[\tilde{\mathbf{U}},\tilde{\mathbf{M}}],\mathbf{M}^k[\tilde{\mathbf{U}},\tilde{\mathbf{M}}])$ denote the unique solution of the linearized step described in Section 5.2, well-posed by Lemma 5.1. Define

$$\mathcal{F}(\tilde{\mathbf{U}},\tilde{\mathbf{M}})\;:=\;\big(\mathbf{U}^k[\tilde{\mathbf{U}},\tilde{\mathbf{M}}],\;\mathbf{M}^k[\tilde{\mathbf{U}},\tilde{\mathbf{M}}]\big). \tag{42}$$

A fixed point of $\mathcal{F}$ is a solution of the scheme.

The product-space norm and absorbing ball

Set

$$\alpha\;:=\;\frac{2\mu_0}{T}\Big(1-\frac{\chi_0}{4}\Big)\;>\;0, \tag{43}$$

which is the relaxation coefficient appearing in the energy estimate (positive under $\chi_0<4$). We work in $\mathcal{Z}_h$ equipped with the norm

$$\|(\mathbf{V},\mathbf{Z})\|_*^2\;:=\;\nu_w\,\|\nabla\mathbf{V}\|^2_{L^2(\Omega)}+\alpha\,\|\mathbf{Z}\|^2_{L^2(\Omega)}+\mu_0\,\|\nabla\Phi(\mathbf{Z})\|^2_{L^2(\Omega)}\,, \tag{44}$$

where $\Phi(\mathbf{Z}) \in \mathcal{X}_h$ is the unique solution of the discrete Poisson problem with magnetization $\mathbf{Z}$ (Section 5.1). All three coefficients are physical parameters independent of the unknown $(\mathbf{V}, \mathbf{Z})$.

The radius R is defined purely from the data at step k, with no dependence on τ:

$$R^2 \;:=\; 2\big(\mathcal{E}^{k-1} + \mathcal{F}^* + 1\big), \qquad \mathcal{F}^* \;:=\; \frac{C_\Omega\,\mu_0}{2\nu_w}\,\Big\|\mathbf{h}_a^k\Big\|^2_{L^2(\Omega)}, \tag{45}$$

where $C_\Omega > 0$ is a Poincaré–Korn constant and $\mathcal{F}^*$ is the τ-independent upper bound on the applied-field forcing ($\tau\,\mathcal{F}^k \le \mathcal{F}^*$ for all $\tau \le 1$). Define

$$B_R \;:=\; \big\{(\tilde{\mathbf{U}}, \tilde{\mathbf{M}}) \in \mathcal{Z}_h : \Big\|(\tilde{\mathbf{U}}, \tilde{\mathbf{M}})\Big\|_* \le R\big\}. \tag{46}$$

Lemma 5.2 (Self-mapping)**.** *There exists*

$$\tau_*(h, R) \;:=\; \big(2\widetilde{C}^2\big)^{-1}\,h^d\,R^{-2}, \tag{47}$$

where $\widetilde{C} > 0$ depends only on C_{inv}, α, μ_0, ν_w, and shape-regularity constants (not on τ, h, or any iterate), such that for $\tau \le \tau_*(h, R)$: $\mathcal{F}(B_R) \subset B_R$.

Proof. Let $(\tilde{\mathbf{U}}, \tilde{\mathbf{M}}) \in B_R$ and $(\mathbf{U}^k, \mathbf{M}^k) = \mathcal{F}(\tilde{\mathbf{U}}, \tilde{\mathbf{M}})$. Test the linearized NS equation with $\mathbf{V} = 2\tau\mathbf{U}^k$ and the linearized magnetization equation with $\mathbf{Z} = 2\tau\mu_0\mathbf{M}^k$, then add.

The coercive terms produce on the left-hand side:

$$\begin{aligned}&\Big\|\mathbf{U}^k\Big\|^2_{L^2(\Omega)} - \Big\|\mathbf{U}^{k-1}\Big\|^2_{L^2(\Omega)} + \Big\|\delta\mathbf{U}^k\Big\|^2_{L^2(\Omega)} + 2\tau\,\nu_w\,\Big\|\nabla\mathbf{U}^k\Big\|^2_{L^2(\Omega)}\\ &+\;\mu_0\Big(\Big\|\mathbf{M}^k\Big\|^2_{L^2(\Omega)} - \Big\|\mathbf{M}^{k-1}\Big\|^2_{L^2(\Omega)} + \Big\|\delta\mathbf{M}^k\Big\|^2_{L^2(\Omega)}\Big) + 2\tau\,\alpha\,\Big\|\mathbf{M}^k\Big\|^2_{L^2(\Omega)} + \tfrac{\mu_0\tau}{T}\,\Big\|\nabla\Phi^k\Big\|^2_{L^2(\Omega)},\end{aligned} \tag{48}$$

using $\nu_\Theta \ge \nu_w > 0$ and the definition (43) of α. The transport, Kelvin, and T2–T3 cancellations from Section 4 hold unchanged for the linearized system (the frozen coefficients $(\tilde{\mathbf{U}}, \tilde{\mathbf{M}})$ do not affect those cancellations). All remaining source terms from previous-step data and forcing are bounded by a constant C_0 depending only on $\mathcal{E}^{k-1}$ and $\mathbf{h}_a^k$.

The only term involving the frozen iterate is the T2 source. Tested with $\mathbf{Z} = 2\tau\mu_0\mathbf{M}^k$ and split via Young's inequality with parameter $\varepsilon = \alpha$:

$$\mu_0\tau\,\Big|\Big((\nabla\times\tilde{\mathbf{U}})\times\tilde{\mathbf{M}},\,\mathbf{M}^k\Big)\Big| \;\le\; \tau\,\alpha\,\Big\|\mathbf{M}^k\Big\|^2_{L^2(\Omega)} + \frac{\tau\,\mu_0^2}{4\alpha}\|\nabla\times\tilde{\mathbf{U}}\|^2_{L^\infty}\,\Big\|\tilde{\mathbf{M}}\Big\|^2_{L^2(\Omega)}. \tag{49}$$

The first term is absorbed exactly into the $2\tau\,\alpha\,\|\mathbf{M}^k\|^2_{L^2(\Omega)}$ term in (48), consuming half of it. No additional condition on T, χ_0, or β is required: absorption is guaranteed by the choice $\varepsilon = \alpha$.

For the second term, the inverse inequality $\|\nabla\times\tilde{\mathbf{U}}\|_{L^\infty} \le C_{\mathrm{inv}}\,h^{-d/2}\,\Big\|\nabla\tilde{\mathbf{U}}\Big\|_{L^2(\Omega)}$ and $(\tilde{\mathbf{U}}, \tilde{\mathbf{M}}) \in B_R$ give:

$$\frac{\tau\,\mu_0^2}{4\alpha}\|\nabla\times\tilde{\mathbf{U}}\|^2_{L^\infty}\,\Big\|\tilde{\mathbf{M}}\Big\|^2_{L^2(\Omega)} \;\le\; \frac{C_{\mathrm{inv}}^2\,\mu_0^2}{4\alpha\,\nu_w\,h^d}\cdot\tau\,R^4 \;\equiv\; \widetilde{C}^2\,h^{-d}\,R^4\,\tau. \tag{50}$$

Collecting (48)–(50) and dropping non-negative squared difference terms:

$$\Big\|(\mathbf{U}^k, \mathbf{M}^k)\Big\|^2_* \;\le\; C_0 + \widetilde{C}^2\,h^{-d}\,R^4\,\tau, \tag{51}$$

where $C_0 \le R^2/2$ by definition (45) of R. The remainder term in (51) satisfies $\widetilde{C}^2 h^{-d} R^4\,\tau \le R^2/2$ *iff* $\tau \le (2\widetilde{C}^2)^{-1}\,h^d\,R^{-2} = \tau_*(h, R)$. Choosing $\tau \le \tau_*(h, R)$ therefore gives $\|(\mathbf{U}^k, \mathbf{M}^k)\|^2_* \le R^2$, i.e. $\mathcal{F}(B_R) \subset B_R$. Since R depends only on the data $(\mathcal{E}^{k-1}, \mathbf{h}_a^k)$ (via (45)) and $\widetilde{C}$ depends only on $(C_{\mathrm{inv}}, \alpha, \mu_0, \nu_w)$ and shape-regularity, the condition $\tau \le \tau_*(h, R)$ is non-circular. □

Lemma 5.3 (Continuity of $\mathcal{F}$)**.** *$\mathcal{F}\colon B_R \to B_R$ is continuous with respect to $\|\cdot\|_*$.*

Proof. Let $(\tilde{\mathbf{U}}^{(n)}, \tilde{\mathbf{M}}^{(n)}) \to (\tilde{\mathbf{U}}, \tilde{\mathbf{M}})$ in $\|\cdot\|_*$. The differences $(e_{\mathbf{U}}, e_{\mathbf{M}}) = \mathcal{F}(\tilde{\mathbf{U}}^{(n)}, \tilde{\mathbf{M}}^{(n)}) - \mathcal{F}(\tilde{\mathbf{U}}, \tilde{\mathbf{M}})$ satisfy the linearized scheme with zero data and T2 source $\frac{1}{2}[(\nabla\times\tilde{\mathbf{U}}^{(n)})\times\tilde{\mathbf{M}}^{(n)} - (\nabla\times\tilde{\mathbf{U}})\times\tilde{\mathbf{M}}] \to \mathbf{0}$ in $[L^2(\Omega)]^d$ (by bilinearity and strong convergence in the finite-dimensional space). The energy estimate for $(e_{\mathbf{U}}, e_{\mathbf{M}})$ with this vanishing source gives $\|(e_{\mathbf{U}}, e_{\mathbf{M}})\|_* \to 0$. □

Theorem 5.4 (Existence)**.** *Under the assumptions of Theorem 4.5 and $\chi_0 < 4$, there exists $\tau_*(h, R) > 0$ given by (47) such that for $\tau \le \tau_*(h, R)$, the scheme has at least one solution at each step $k = 1, \dots, K$.*

Proof. Since $\mathcal{U}_h$ and $\mathcal{M}_h$ are finite-dimensional vector spaces, the product $\mathcal{Z}_h$ is a finite-dimensional Euclidean space and Brouwer's theorem applies. B_R is a convex compact subset of this space. By Lemmas 5.2–5.3, $\mathcal{F}\colon B_R \to B_R$ is continuous with respect to $\|\cdot\|_*$. The Brouwer Fixed Point Theorem [5] yields a fixed point $(\mathbf{U}^k, \mathbf{M}^k) \in B_R$, which together with $(\Theta^k, \Psi^k, \Phi^k, P^k)$ determined as in Section 5.1 constitutes a full solution of the scheme. □

5.4 Conditional Uniqueness

Theorem 5.5 (Uniqueness)**.** *Under the assumptions of Theorem 4.5, there exists $\tau_0 = \tau_0(h, R) > 0$ such that for $\tau < \tau_0$ the solution of the scheme is unique at each step k.*

Proof. Let $(\mathbf{U}_1^k, \mathbf{M}_1^k)$ and $(\mathbf{U}_2^k, \mathbf{M}_2^k)$ be two solutions of the scheme at step k, both in B_R. Set $e_{\mathbf{U}} = \mathbf{U}_1^k - \mathbf{U}_2^k$, $e_{\mathbf{M}} = \mathbf{M}_1^k - \mathbf{M}_2^k$, $e_\Phi = \Phi_1^k - \Phi_2^k$, and similarly for the remaining variables. These differences satisfy a linear system with zero previous-step data. All terms that are linear in the current unknowns the time-derivative, viscous, relaxation, Poisson, and Kelvin terms cancel exactly between the two solution equations, leaving only the T2 nonlinear difference (52) on the right-hand side.

$$(\nabla \times \mathbf{U}_1^k) \times \mathbf{M}_1^k - (\nabla \times \mathbf{U}_2^k) \times \mathbf{M}_2^k = (\nabla \times e_{\mathbf{U}}) \times \mathbf{M}_1^k + (\nabla \times \mathbf{U}_2^k) \times e_{\mathbf{M}}. \tag{52}$$

Test the difference magnetization equation with $\mathbf{Z} = 2\tau\mu_0 e_{\mathbf{M}}$ and the difference NS equation with $\mathbf{V} = 2\tau e_{\mathbf{U}}$.

The coercive terms give on the left-hand side:

$$\frac{1}{\tau}\,|\!|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!|\!|^2 + 2\tau\,\|\sqrt{\nu_\Theta}\,\mathbf{T}(e_{\mathbf{U}})\|^2_{L^2(\Omega)} + \frac{2\mu_0\tau}{T}\Big(1 - \frac{\chi_0}{4}\Big)\,\|e_{\mathbf{M}}\|^2_{L^2(\Omega)}\,. \tag{53}$$

The T3 difference and Kelvin force difference cancel exactly by the same mechanism as in Theorem 4.5 (Steps 5' and T2–T3 cross-cancellation), applied to the difference equations.

For the T2 difference (52):

Second term: By Lemma 4.4(24) with $\mathbf{c} = \nabla \times \mathbf{U}_2^k$ and $\mathbf{a} = e_{\mathbf{M}}$: $\big((\nabla \times \mathbf{U}_2^k) \times e_{\mathbf{M}},\, e_{\mathbf{M}}\big) = 0$.

First term: Using the trilinearity bound (**??**) and $\|\mathbf{M}_1^k\|_{L^2(\Omega)} \le \mu_0^{-1/2} R$:

$$\begin{aligned}\mu_0\tau\Big|\Big((\nabla \times e_{\mathbf{U}}) \times \mathbf{M}_1^k,\, e_{\mathbf{M}}\Big)\Big| \;&\le\; C_{\mathrm{inv}}\mu_0\tau\, h^{-1}\,\mu_0^{-1/2} R\,\|e_{\mathbf{U}}\|_{L^2(\Omega)}\,\|e_{\mathbf{M}}\|_{L^2(\Omega)} \\ &\le\; \frac{C_*^2\tau^2}{2}\,|\!|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!|\!|^2 + \frac{1}{2\tau}\,|\!|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!|\!|^2\,,\end{aligned} \tag{54}$$

where $C_* = C_{\mathrm{inv}}\mu_0^{1/2} h^{-1} R$ and the last step uses Young's inequality with parameter τ.

Substituting (54) into (53):

$$\frac{1}{2\tau}\,|\!|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!|\!|^2 \;\le\; \frac{C_*^2\tau^2}{2}\,|\!|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!|\!|^2\,, \tag{55}$$

which rearranges to $\|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!\|^2 \left(\frac{1}{2\tau} - \frac{C_*^2\tau^2}{2}\right) \le 0$. Setting

$$\tau_0 \;:=\; C_*^{-2/3} \tag{56}$$

ensures $\frac{1}{2\tau} > \frac{C_*^2\tau^2}{2}$ for $\tau < \tau_0$, so $\|\!|(e_{\mathbf{U}}, e_{\mathbf{M}})|\!\| = 0$. Uniqueness of the remaining differences $(e_\Theta, e_\Psi, e_\Phi, e_P)$ then follows from the same energy argument as in Lemma 5.1 applied to the difference equations. □

Remark 5.6 (Stability vs. existence vs. uniqueness)**.** Theorem 4.5 establishes *unconditional* energy stability: it holds for all $\tau > 0$ and all $h > 0$. Existence and uniqueness, however, both require time-step restrictions because the nonlinear T2 term couples $(\mathbf{U}^k, \mathbf{M}^k)$ at the current time level.

- *Existence* (Theorem 5.4) requires $\tau \le \tau_*(h, R) = (2\widetilde{C}^2)^{-1}\, h^d\, R^{-2}$, arising from the absorbing-ball self-mapping argument. In two dimensions this is a parabolic-type CFL, $\tau_* \sim h^2/R^2$, which is the standard restriction one expects for a fully nonlinear Brouwer fixed-point construction.
- *Uniqueness* (Theorem 5.5) requires $\tau < \tau_0 = C_*^{-2/3} \sim h^{2/3}/R^{2/3}$, arising from a contractivity argument for the difference of two solutions.

For typical data with $R \ge \sqrt{2}$ and $h \ll R$, one has $\tau_* \ll \tau_0$, so the existence-CFL is the binding constraint in practice; whenever existence is achieved, uniqueness is automatic.

Both restrictions depend on the mesh and the energy level, and are typically far below accuracy-driven time steps.

Remark 5.7 (Comparison with related schemes)**.** Nochetto et al. [2] (T1, T2, T3 absent) establish existence and uniqueness for their scheme via coercivity of the underlying bilinear form, which is purely linear. Zhang et al. [3] (T1, T2, T3 present) maintain a fully decoupled linear structure at each step via auxiliary intermediate variables, so their existence proof is also purely linear. The present scheme is the first in this family to introduce a genuinely nonlinear coupling at the current time level (T2 with $\mathbf{U}^k$ and $\mathbf{M}^k$), for which Brouwer's theorem is the natural tool.

6 Convergence

We prove that as $h, \tau \to 0$, subsequences of the discrete solutions produced by the scheme converge to a weak solution of the continuous ferrohydrodynamic system. The argument proceeds in four steps: uniform bounds, time-derivative bounds, compactness, and passage to the limit. Throughout this section we assume the conditions of Theorems 4.5 and 5.5, $\chi_0 < 4$, $\beta T \le 1$, and $\mathbf{h}_a \in L^2(0, T; [H^1(\Omega)]^d)$.

Assumption 6.1 (Spatial integrability of $\mathbf{M}_{h,\tau}$)**.** We assume that the discrete magnetization satisfies a uniform $L^4_{t,x}$ bound:

$$\sup_{(h,\tau)} \|\bar{\mathbf{M}}_{h,\tau}\|_{L^4(0,T;\, L^4(\Omega)^d)} \;\le\; C, \tag{57}$$

with C independent of h and τ but depending on the data $(\mathcal{E}^0, \mathbf{h}_a)$. This regularity is not implied by the discrete energy law (27) for the DG space $\mathcal{M}_h \subset [L^2(\Omega)]^d$ (which only controls $\bar{\mathbf{M}}_{h,\tau}$ in $L^\infty(0, T; L^2(\Omega)^d)$), but is needed to obtain a uniform-in-h bound on $\partial_t \hat{\mathbf{M}}_{h,\tau}$ in a negative-order space and hence to apply the Aubin–Lions theorem to the magnetization. In two dimensions, (57) can equivalently be enforced by adding a small $\eta\, \|\nabla \mathbf{M}^k\|^2_{L^2}$ stabilization term to the Cahn–Hilliard–magnetization coupling at the cost of modifying the consistency analysis; we work directly under Assumption 6.1 to keep the scheme as stated in Section 3.

6.1 Piecewise-Constant and Piecewise-Linear Extensions

Given discrete solutions $\{(\Theta^k, \mathbf{M}^k, \mathbf{U}^k, P^k)\}_{k=0}^K$ at time nodes $t_k = k\tau$, define piecewise-constant and piecewise-linear extensions to $[0, T]$:

$$\bar{\mathbf{U}}_{h,\tau}(t) := \mathbf{U}^k, \quad \bar{\mathbf{M}}_{h,\tau}(t) := \mathbf{M}^k, \quad \bar{\Theta}_{h,\tau}(t) := \Theta^k, \quad \bar{P}_{h,\tau}(t) := P^k \quad \text{for } t \in (t_{k-1}, t_k], \tag{58}$$

$$\hat{\mathbf{U}}_{h,\tau}(t) := \tfrac{t_k - t}{\tau}\mathbf{U}^{k-1} + \tfrac{t - t_{k-1}}{\tau}\mathbf{U}^k, \quad \hat{\mathbf{M}}_{h,\tau}(t) := \tfrac{t_k - t}{\tau}\mathbf{M}^{k-1} + \tfrac{t - t_{k-1}}{\tau}\mathbf{M}^k, \tag{59}$$

for $t \in [t_{k-1}, t_k]$, and similarly for $\hat{\Theta}_{h,\tau}$. The piecewise-linear functions satisfy $\partial_t \hat{\mathbf{U}}_{h,\tau} = \delta\mathbf{U}^k/\tau$ a.e. on (t_{k-1}, t_k).

6.2 Uniform A Priori Bounds

Proposition 6.2 (Uniform bounds)**.** *There exists $C = C(T, \mathcal{E}^0, \mathbf{h}_a) > 0$, independent of h and τ, such that:*

$$\|\bar{\mathbf{U}}_{h,\tau}\|_{L^\infty(0,T;\,[L^2(\Omega)]^d)} + \|\bar{\mathbf{U}}_{h,\tau}\|_{L^2(0,T;\,[H_0^1(\Omega)]^d)} \le C, \tag{60}$$

$$\|\bar{\mathbf{M}}_{h,\tau}\|_{L^\infty(0,T;\,[L^2(\Omega)]^d)} \le C, \tag{61}$$

$$\|\nabla\bar{\Phi}_{h,\tau}\|_{L^\infty(0,T;\,[L^2(\Omega)]^d)} \le C, \tag{62}$$

$$\|\bar{\Theta}_{h,\tau}\|_{L^\infty(0,T;\,H^1(\Omega))} \le C, \tag{63}$$

$$\|\bar{\Psi}_{h,\tau}\|_{L^2(0,T;\,H^1(\Omega))} \le C. \tag{64}$$

Proof. Summing (27) over $k = 1, \ldots, K$ and using $K\tau = T$: the $L^\infty(L^2)$ bound on $\mathbf{U}$ follows from $\mathcal{E}^k \ge \frac{1}{2}\left\|\mathbf{U}^k\right\|_{L^2(\Omega)}^2$; the $L^2(H^1)$ bound from $\sum_k \tau\mathcal{D}_p^k \ge c_\nu \sum_k \tau \left\|\nabla\mathbf{U}^k\right\|_{L^2(\Omega)}^2$ with $c_\nu > 0$ depending on ν_w and Korn's inequality; the $L^\infty(L^2)$ bound on $\mathbf{M}$ from $\mathcal{E}^k \ge \frac{\mu_0}{2}\left\|\mathbf{M}^k\right\|_{L^2(\Omega)}^2$; the $L^\infty(L^2)$ bound on $\nabla\Phi$ from $\mathcal{E}^k \ge \frac{\mu_0}{2}\left\|\nabla\Phi^k\right\|_{L^2(\Omega)}^2$; the $L^\infty(H^1)$ bound on Θ from $\mathcal{E}^k \ge \frac{\lambda}{2}\left\|\nabla\Theta^k\right\|_{L^2(\Omega)}^2$; and the $L^2(H^1)$ bound on Ψ from $\sum_k \tau\mathcal{D}_p^k \ge \frac{\lambda\gamma}{\varepsilon}\sum_k \tau \left\|\nabla\Psi^k\right\|_{L^2(\Omega)}^2$. □

6.3 Time-Derivative Bounds

To apply the Aubin–Lions theorem we need uniform bounds on the time derivatives of the piecewise-linear extensions in negative-order spaces.

Lemma 6.3 (Time-derivative bounds)**.** *Under the conditions of Proposition 6.2:*

$$\|\partial_t\hat{\mathbf{U}}_{h,\tau}\|_{L^2(0,T;\,[H^{-1}(\Omega)]^d)} \le C, \tag{65}$$

$$\|\partial_t\hat{\mathbf{M}}_{h,\tau}\|_{L^2(0,T;\,[H^{-1}(\Omega)]^d)} \le C, \tag{66}$$

$$\|\partial_t\hat{\Theta}_{h,\tau}\|_{L^2(0,T;\,H^{-1}(\Omega))} \le C. \tag{67}$$

Proof. Bound (65)*.* For any $\boldsymbol{\phi} \in [H_0^1(\Omega)]^d$ with $\|\nabla\boldsymbol{\phi}\|_{L^2(\Omega)} \le 1$, let $\mathbf{V} = \mathcal{P}_h\boldsymbol{\phi}$ be the H^1-projection onto $\mathcal{U}_h$, which satisfies $\|\nabla\mathbf{V}\|_{L^2(\Omega)} \le C$ uniformly. Test (15b) with $\mathbf{V}$:

$$\begin{aligned}\left|\Big(\delta\mathbf{U}^k/\tau,\, \mathbf{V}\Big)\right| \le \left|B_h(\mathbf{U}^{k-1}, \mathbf{U}^k, \mathbf{V})\right| + \left|\Big(\nu_\Theta\mathbf{T}(\mathbf{U}^k),\, \mathbf{T}(\mathbf{V})\Big)\right| + \left|\Big(P^k,\, \nabla\cdot\mathbf{V}\Big)\right| \\ + \mu_0\left|B_h^m(\mathbf{V}, \mathbf{H}^k, \mathbf{M}^k)\right| + \tfrac{\lambda}{\varepsilon}\left|\Big(\Theta^{k-1}\nabla\Psi^k,\, \mathbf{V}\Big)\right| + |\mathrm{T3}|.\end{aligned}$$

Each term on the right is bounded using: the skew-symmetry of B_h and $\left\|\mathbf{U}^{k-1}\right\|_{L^2(\Omega)} \le C$ for the advection; $\left\|\nabla\mathbf{U}^k\right\|_{L^2(\Omega)} \le C$ for viscosity; the LBB condition and $\left\|\mathbf{U}^k\right\|_{L^2(\Omega)} \le C$ to bound the pressure via $\left\|P^k\right\|_{L^2(\Omega)} \le C$ (see (69) below); the trilinearity bound (**??**) and $\left\|\mathbf{M}^k\right\|_{L^2(\Omega)}, \left\|\nabla\Phi^k\right\|_{L^2(\Omega)} \le C$ for Kelvin; $\left\|\Theta^{k-1}\right\|_{L^2(\Omega)} \le C$ and $\left\|\nabla\Psi^k\right\|_{L^2(\Omega)} \le C$ for the

Cahn–Hilliard force; and $\left\|\mathbf{M}^k\right\|_{L^2(\Omega)}, \left\|\nabla\Phi^k\right\|_{L^2(\Omega)} \le C$ for T3. Squaring, summing over k, and multiplying by τ gives (65).

Bound (66). Test (15a) with $\mathbf{Z} = \mathcal{P}_h\boldsymbol{\psi}$ for $\boldsymbol{\psi} \in [H^1(\Omega)]^d$, $\|\boldsymbol{\psi}\|_{H^1} \le 1$:

$$\begin{aligned}\left|\Big(\delta\mathbf{M}^k/\tau,\, \mathbf{Z}\Big)\right| &\le \left|B_h^m(\mathbf{U}^k, \mathbf{Z}, \mathbf{M}^k)\right| + \tfrac{1}{T}\left|\Big(\mathbf{M}^k - \tilde{\chi}_\Theta\mathbf{H}^k,\, \mathbf{Z}\Big)\right| \\ &\quad + \beta\left|\Big(\mathbf{M}^{k-1} \times (\mathbf{M}^{k-1} \times \mathbf{H}^k),\, \mathbf{Z}\Big)\right| + \tfrac{1}{2}\left|\Big((\nabla \times \mathbf{U}^k) \times \mathbf{M}^k,\, \mathbf{Z}\Big)\right|.\end{aligned}$$

Each term is bounded using the trilinearity of B_h^m, the uniform bounds on $\mathbf{M}^k$, $\mathbf{H}^k$, and $\mathbf{U}^k$ from Proposition 6.2. The T2 term requires more care. In two dimensions, the Sobolev embedding $H^1(\Omega) \hookrightarrow L^4(\Omega)$ gives $\|\mathbf{Z}\|_{L^4} \le C\|\mathbf{Z}\|_{H^1} \le C$, so by Hölder's inequality with the triple $\frac{1}{2}+\frac{1}{4}+\frac{1}{4} = 1$:

$$\left|\Big((\nabla \times \mathbf{U}^k) \times \mathbf{M}^k,\, \mathbf{Z}\Big)\right| \;\le\; \|\nabla\mathbf{U}^k\|_{L^2}\,\|\mathbf{M}^k\|_{L^4}\,\|\mathbf{Z}\|_{L^4} \;\le\; C\,\|\nabla\mathbf{U}^k\|_{L^2}\,\|\mathbf{M}^k\|_{L^4}. \tag{68}$$

Squaring (68), summing over k, and multiplying by τ gives a contribution bounded by $C\,\|\bar{\mathbf{U}}\|^2_{L^2(0,T;H^1)}\,\|\bar{\mathbf{M}}_{h,\tau}\|^2_{L^4(}$ which is uniformly bounded by Proposition 6.2 and Assumption 6.1. Summing in k gives (66).

Bound (67). Test (14a) with $\Lambda = \mathcal{P}_h\phi$ for $\phi \in H^1(\Omega)$, $\|\phi\|_{H^1} \le 1$:

$$\begin{aligned}\left|\Big(\delta\Theta^k/\tau,\, \Lambda\Big)\right| &\le \left|\Big(\mathbf{U}^{k-1}\Theta^{k-1},\, \nabla\Lambda\Big)\right| + \gamma\left|\Big(\nabla\Psi^k,\, \nabla\Lambda\Big)\right| \\ &\le \left\|\mathbf{U}^{k-1}\right\|_{L^2(\Omega)}\left\|\Theta^{k-1}\right\|_{L^2(\Omega)}\|\nabla\Lambda\|_{L^2(\Omega)} + \gamma\left\|\nabla\Psi^k\right\|_{L^2(\Omega)}\|\nabla\Lambda\|_{L^2(\Omega)}.\end{aligned}$$

Using $\left\|\mathbf{U}^{k-1}\right\|_{L^2(\Omega)}, \left\|\Theta^{k-1}\right\|_{L^2(\Omega)}, \left\|\nabla\Psi^k\right\|_{L^2(\Omega)} \le C$ and summing gives (67). □

Lemma 6.4 (Pressure bound)**.**

$$\|\bar{P}_{h,\tau}\|_{L^2(0,T;\,L^2(\Omega))} \;\le\; C. \tag{69}$$

Proof. The LBB condition for $(\mathcal{U}_h, \mathcal{P}_h)$ gives, at each step k:

$$\beta_0\left\|P^k\right\|_{L^2(\Omega)} \;\le\; \sup_{\mathbf{V}\in\mathcal{U}_h\setminus\{0\}} \frac{\big(P^k,\, \nabla\cdot\mathbf{V}\big)}{\|\nabla\mathbf{V}\|_{L^2(\Omega)}}.$$

Extracting $\big(P^k,\, \nabla\cdot\mathbf{V}\big)$ from (15b) and bounding the right-hand side using Proposition 6.2 yields $\left\|P^k\right\|_{L^2(\Omega)} \le C$ for each k. Multiplying by τ and summing gives (69). □

6.4 Compactness

Proposition 6.5 (Weak and strong limits)**.** *There exists a subsequence $\{h_n, \tau_n\} \to 0$ (not relabelled) and limit functions $\mathbf{U} \in L^2(0,T;[H_0^1(\Omega)]^d)$, $\mathbf{M} \in L^2(0,T;[L^2(\Omega)]^d)$, $\nabla\Phi \in L^2(0,T;[L^2(\Omega)]^d)$, $\Theta \in L^\infty(0,T;H^1(\Omega))$, $P \in L^2(0,T;L^2(\Omega))$, such that:*

$$\bar{\mathbf{U}}_{h,\tau} \rightharpoonup \mathbf{U} \qquad \textit{weakly in } L^2(0,T;[H_0^1(\Omega)]^d), \tag{70}$$

$$\bar{\mathbf{M}}_{h,\tau} \rightharpoonup \mathbf{M} \qquad \textit{weakly in } L^2(0,T;[L^2(\Omega)]^d), \tag{71}$$

$$\nabla\bar{\Phi}_{h,\tau} \rightharpoonup \nabla\Phi \qquad \textit{weakly in } L^2(0,T;[L^2(\Omega)]^d), \tag{72}$$

The weak convergence (72) is a consequence of (71): since the discrete Poisson operator $\mathbf{M}^k \mapsto \nabla\Phi^k$ is the solution map of $-\Delta\Phi^k = -\nabla\cdot\mathbf{M}^k$ (with $\mathbf{h}_a^k$ contribution), which is linear and continuous from $[L^2(\Omega)]^d$ to $[H^1(\Omega)]^d$ by elliptic regularity, weak convergence of $\bar{\mathbf{M}}_{h,\tau}$ in $L^2(L^2)$ implies weak convergence of $\nabla\bar{\Phi}_{h,\tau}$ in $L^2(H^1)$, and hence in $L^2(L^2)$. The limit field $\mathbf{H} := \nabla\Phi$ therefore satisfies $\mathbf{H} \in L^2(0,T;[L^2(\Omega)]^d)$ with $\|\mathbf{H}\|_{L^2(\Omega)} \lesssim \|\mathbf{M}\|_{L^2(\Omega)}$ pointwise in time.

$$\bar{\Theta}_{h,\tau} \xrightarrow{*} \Theta \qquad \textit{weak-}* \textit{ in } L^\infty(0,T;H^1(\Omega)), \tag{73}$$

$$\bar{P}_{h,\tau} \rightharpoonup P \qquad \textit{weakly in } L^2(0,T;L^2(\Omega)), \tag{74}$$

and the following strong convergences hold along a further subsequence:

$$\bar{\mathbf{U}}_{h,\tau} \to \mathbf{U} \qquad \textit{strongly in } L^2(0,T;[L^2(\Omega)]^d), \tag{75}$$

$$\bar{\mathbf{M}}_{h,\tau} \to \mathbf{M} \qquad \textit{strongly in } L^2(0,T;[L^2(\Omega)]^d), \tag{76}$$

$$\bar{\Theta}_{h,\tau} \to \Theta \qquad \textit{strongly in } L^2(0,T;L^2(\Omega)). \tag{77}$$

Proof. The weak convergences (70)–(74) follow from Proposition 6.2, Lemma 6.4, and the Banach–Alaoglu theorem, noting that $L^2(H^1)$ and $L^2(L^2)$ are reflexive.

For strong convergence, we apply the Aubin–Lions theorem (see [5]). Consider the triple $H_0^1(\Omega) \subset\subset L^2(\Omega) \subset H^{-1}(\Omega)$, where the first embedding is compact. Proposition 6.2 gives $\hat{\mathbf{U}}_{h,\tau}$ bounded in $L^2(0,T;H_0^1(\Omega))$, and Lemma 6.3(65) gives $\partial_t \hat{\mathbf{U}}_{h,\tau}$ bounded in $L^2(0,T;H^{-1}(\Omega))$. Aubin–Lions yields (75) along a subsequence.

For $\mathbf{M}$: Proposition 6.2 gives $\hat{\mathbf{M}}_{h,\tau}$ bounded in $L^\infty(0,T;L^2(\Omega))$ (hence in $L^2(0,T;L^2(\Omega))$). Under Assumption 6.1, the bound on $\partial_t \hat{\mathbf{M}}_{h,\tau}$ derived in Lemma 6.3(66) from (68) is uniform in h, since $\|\bar{\mathbf{M}}_{h,\tau}\|_{L^4(L^4)} \le C$ controls the only nonstandard factor in the T2 estimate. Hence $\partial_t \hat{\mathbf{M}}_{h,\tau}$ is uniformly bounded in $L^2(0,T;H^{-1}(\Omega))$. The triple $L^2(\Omega) \subset\subset L^2(\Omega)$ collapses, so we use instead the triple $L^2(\Omega) \subset\subset H^{-\delta}(\Omega) \subset H^{-1}(\Omega)$ for any $\delta \in (0,1)$, with $L^2 \hookrightarrow H^{-\delta}$ compact (Rellich). Aubin–Lions gives strong convergence in $L^2(0,T;H^{-\delta}(\Omega))$; combined with the $L^\infty(L^2)$ bound, interpolation gives $\bar{\mathbf{M}}_{h,\tau} \to \mathbf{M}$ strongly in $L^2(0,T;L^2(\Omega))$, establishing (76).

Bound (77) follows by the same argument applied to $\hat{\Theta}_{h,\tau}$, using Proposition 6.2(63) and Lemma 6.3(67). □

6.5 Identification of Nonlinear Limits

The strong convergences of Proposition 6.5 are the key tool for identifying limits of all nonlinear terms.

Lemma 6.6 (Nonlinear limit identification)**.** *The following weak limits hold in $L^1(0,T;[L^1(\Omega)]^d)$:*

$$\bar{\mathbf{U}}_{h,\tau} \otimes \bar{\mathbf{U}}_{h,\tau} \rightharpoonup \mathbf{U} \otimes \mathbf{U}, \tag{78}$$

$$\bar{\mathbf{U}}_{h,\tau}\, \bar{\Theta}_{h,\tau} \rightharpoonup \mathbf{U}\Theta, \tag{79}$$

$$(\nabla \times \bar{\mathbf{U}}_{h,\tau}) \times \bar{\mathbf{M}}_{h,\tau} \rightharpoonup (\nabla \times \mathbf{U}) \times \mathbf{M}, \tag{80}$$

$$\bar{\mathbf{M}}_{h,\tau} \times \nabla \bar{\Phi}_{h,\tau} \rightharpoonup \mathbf{M} \times \nabla \Phi, \tag{81}$$

$$\bar{\mathbf{M}}_{h,\tau} \times (\bar{\mathbf{M}}_{h,\tau} \times \nabla \bar{\Phi}_{h,\tau}) \rightharpoonup \mathbf{M} \times (\mathbf{M} \times \nabla \Phi). \tag{82}$$

Proof. Throughout, we use the principle that strong $\times$ weak $\to$ weak in L^1: if $f_n \to f$ strongly in L^2 and $g_n \rightharpoonup g$ weakly in L^2, then $f_n g_n \rightharpoonup fg$ weakly in L^1.

Limit (78): $\bar{\mathbf{U}}_{h,\tau} \otimes \bar{\mathbf{U}}_{h,\tau} - \mathbf{U} \otimes \mathbf{U} = (\bar{\mathbf{U}}_{h,\tau} - \mathbf{U}) \otimes \bar{\mathbf{U}}_{h,\tau} + \mathbf{U} \otimes (\bar{\mathbf{U}}_{h,\tau} - \mathbf{U})$. Both terms tend to zero in L^1 since $\bar{\mathbf{U}}_{h,\tau} \to \mathbf{U}$ strongly in $L^2(L^2)$ and $\bar{\mathbf{U}}_{h,\tau}$ is bounded in $L^2(L^2)$.

Limit (79): By (75) and (77), both factors converge strongly in $L^2(L^2)$, so their product converges strongly in $L^1(L^1)$.

Limit (80): Write $(\nabla \times \bar{\mathbf{U}}_{h,\tau}) \times \bar{\mathbf{M}}_{h,\tau} - (\nabla \times \mathbf{U}) \times \mathbf{M} = (\nabla \times \bar{\mathbf{U}}_{h,\tau}) \times (\bar{\mathbf{M}}_{h,\tau} - \mathbf{M}) + (\nabla \times (\bar{\mathbf{U}}_{h,\tau} - \mathbf{U})) \times \mathbf{M}$. For the first term: $\nabla \times \bar{\mathbf{U}}_{h,\tau}$ is bounded in $L^2(L^2)$ and $\bar{\mathbf{M}}_{h,\tau} - \mathbf{M} \to 0$ strongly in $L^2(L^2)$ by (76), so the term tends to zero in $L^1(L^1)$ by strong $\times$ bounded convergence. For the second: $\mathbf{M} \in L^2(L^2)$ is fixed and $\nabla \times (\bar{\mathbf{U}}_{h,\tau} - \mathbf{U}) \rightharpoonup 0$ weakly in $L^2(L^2)$ by (70), so $(\nabla \times (\bar{\mathbf{U}}_{h,\tau} - \mathbf{U})) \times \mathbf{M} \rightharpoonup 0$ weakly in $L^1(L^1)$ by the strong $\times$ weak principle.

Limit (81): $\bar{\mathbf{M}}_{h,\tau} \times \nabla \bar{\Phi}_{h,\tau} - \mathbf{M} \times \nabla \Phi = (\bar{\mathbf{M}}_{h,\tau} - \mathbf{M}) \times \nabla \bar{\Phi}_{h,\tau} + \mathbf{M} \times (\nabla \bar{\Phi}_{h,\tau} - \nabla \Phi)$. First term: $\bar{\mathbf{M}}_{h,\tau} - \mathbf{M} \to 0$ strongly in $L^2(L^2)$ by (76), and $\nabla \bar{\Phi}_{h,\tau}$ is bounded in $L^2(L^2)$, so the term vanishes in L^1. Second term: $\mathbf{M} \in L^2(L^2)$ and $\nabla \bar{\Phi}_{h,\tau} - \nabla \Phi \rightharpoonup 0$ weakly in $L^2(L^2)$ by (72), so the product $\mathbf{M} \times (\nabla \bar{\Phi}_{h,\tau} - \nabla \Phi) \rightharpoonup 0$ weakly in $L^1(L^1)$ by the strong–weak principle applied with $\mathbf{M}$ fixed.

Note that both $\bar{\mathbf{M}}_{h,\tau}$ and $\nabla\bar{\Phi}_{h,\tau}$ converge only weakly in L^2; the first term above is the only one where the strong $\times$ bounded mechanism is essential. In the cross-product $\mathbf{M}_h \times \mathbf{H}_h$ that appears in the magnetic stress and Kelvin force, an alternative approach is to use the div–curl lemma [4]: since $\nabla \times \mathbf{H}_h = \nabla \times \nabla\Phi_h = \mathbf{0}$ (curl of a gradient) and $\nabla \cdot \mathbf{M}_h$ is controlled by the Poisson equation, the pair $(\mathbf{M}_h, \mathbf{H}_h)$ satisfies the hypotheses of the div–curl lemma, which yields $\mathbf{M}_h \cdot \mathbf{H}_h \to \mathbf{M} \cdot \mathbf{H}$ strongly in L^1 (and hence $\mathbf{M}_h \times \mathbf{H}_h \rightharpoonup \mathbf{M} \times \mathbf{H}$ in L^1 via the vector identity $\mathbf{M} \times (\mathbf{M} \times \mathbf{H}) = (\mathbf{M} \cdot \mathbf{H})\mathbf{M} - |\mathbf{M}|^2\mathbf{H}$). The decomposition used above for T3 and T1 is equivalent and avoids invoking div–curl directly.

Limit (82)*:* By (81), $\bar{\mathbf{M}}_{h,\tau} \times \nabla\bar{\Phi}_{h,\tau} \rightharpoonup \mathbf{M} \times \nabla\Phi$ weakly in L^1. Write $\bar{\mathbf{M}}_{h,\tau} \times (\bar{\mathbf{M}}_{h,\tau} \times \nabla\bar{\Phi}_{h,\tau}) - \mathbf{M} \times (\mathbf{M} \times \nabla\Phi) = (\bar{\mathbf{M}}_{h,\tau} - \mathbf{M}) \times (\bar{\mathbf{M}}_{h,\tau} \times \nabla\bar{\Phi}_{h,\tau}) + \mathbf{M} \times [(\bar{\mathbf{M}}_{h,\tau} \times \nabla\bar{\Phi}_{h,\tau}) - (\mathbf{M} \times \nabla\Phi)]$. The first term: $\bar{\mathbf{M}}_{h,\tau} - \mathbf{M} \to 0$ strongly in $L^2(L^2)$ and $\bar{\mathbf{M}}_{h,\tau} \times \nabla\bar{\Phi}_{h,\tau}$ is bounded in L^1 (by $\left\|\mathbf{M}^k\right\|_{L^2(\Omega)} \left\|\nabla\Phi^k\right\|_{L^2(\Omega)} \leq C$), so the term vanishes in L^1. The second term: $\mathbf{M} \in L^2(L^2)$ and the bracket converges weakly to zero in L^1 by (81), giving weak L^1 convergence of the product. □

Vanishing of DG interior face terms

The DG transport operator B_h^m contains interior face terms involving jumps $[\![\mathbf{M}_{h,\tau}]\!]$. For any smooth test function $\mathbf{Z} \in [C_c^\infty(\Omega)]^d$, the face contribution is:

$$\sum_{F \in \mathcal{F}^i} \int_F (\bar{\mathbf{U}}_{h,\tau} \cdot \mathbf{n}^-)^- \, [\![\bar{\mathbf{M}}_{h,\tau}]\!] \cdot \mathbf{Z} \, \mathrm{d}s.$$

Since $\mathbf{Z}$ is smooth, $|\mathbf{Z}|_F \leq C$ uniformly on each face. The face jump term satisfies $h^{1/2}\|[\![\mathbf{M}^k]\!]\|_{L^2(\mathcal{F}^i)} \leq C \left\|\mathbf{M}^k\right\|_{L^2(\Omega)} \leq C$ by a standard DG trace inequality. Hence the face contribution is bounded by $Ch^{1/2} \left\|\mathbf{U}^{k-1}\right\|_{L^2(\Omega)} \to 0$ as $h \to 0$, so $B_h^m(\bar{\mathbf{U}}, \mathbf{Z}, \bar{\mathbf{M}}) \to \int_\Omega (\bar{\mathbf{U}} \cdot \nabla\mathbf{Z}) \cdot \bar{\mathbf{M}} \, \mathrm{d}x$ in the limit.

6.6 The Continuous Weak Formulation

The *weak solution* $(\mathbf{U}, \mathbf{M}, \Phi, \Theta, \Psi, P)$ satisfies, for all $\mathbf{V} \in [C_c^\infty(\Omega \times [0,T))]^d$, $\mathbf{Z} \in [C_c^\infty(\Omega \times [0,T))]^d$, $X, \Lambda, \Upsilon \in C_c^\infty(\Omega \times [0,T))$, $Q \in C_c^\infty(\Omega \times [0,T))$:

$$-\int_0^T (\mathbf{U},\, \partial_t \mathbf{V})\, \mathrm{d}t + \int_0^T ((\mathbf{U}\cdot\nabla)\mathbf{U},\, \mathbf{V})\, \mathrm{d}t + \int_0^T (\nu_\Theta \mathbf{T}(\mathbf{U}),\, \mathbf{T}(\mathbf{V}))\, \mathrm{d}t - \int_0^T (P,\, \nabla\cdot\mathbf{V})\, \mathrm{d}t$$
$$= \mu_0 \int_0^T ((\mathbf{M}\cdot\nabla)\mathbf{H},\, \mathbf{V})\, \mathrm{d}t + \frac{\mu_0}{2}\int_0^T (\mathbf{M}\times\mathbf{H},\, \nabla\times\mathbf{V})\, \mathrm{d}t + \frac{\lambda}{\varepsilon}\int_0^T (\Theta\nabla\Psi,\, \mathbf{V})\, \mathrm{d}t, \tag{83a}$$

$$-\int_0^T (\mathbf{M},\, \partial_t \mathbf{Z})\, \mathrm{d}t + \int_0^T ((\mathbf{U}\cdot\nabla)\mathbf{M},\, \mathbf{Z})\, \mathrm{d}t + \frac{1}{T}\int_0^T (\mathbf{M} - \tilde{\chi}_\Theta \mathbf{H},\, \mathbf{Z})\, \mathrm{d}t$$
$$-\beta\int_0^T (\mathbf{M}\times(\mathbf{M}\times\mathbf{H}),\, \mathbf{Z})\, \mathrm{d}t + \frac{1}{2}\int_0^T ((\nabla\times\mathbf{U})\times\mathbf{M},\, \mathbf{Z})\, \mathrm{d}t = 0, \tag{83b}$$

$$\int_0^T (\nabla\Phi,\, \nabla X)\, \mathrm{d}t + \int_0^T (\mathbf{M},\, \nabla X)\, \mathrm{d}t = \int_0^T (\mathbf{h}_a,\, \nabla X)\, \mathrm{d}t, \tag{83c}$$

$$-\int_0^T (\Theta,\, \partial_t \Lambda)\, \mathrm{d}t - \int_0^T (\mathbf{U}\Theta,\, \nabla\Lambda)\, \mathrm{d}t + \gamma\int_0^T (\nabla\Psi,\, \nabla\Lambda)\, \mathrm{d}t = 0, \tag{83d}$$

$$\int_0^T (\Psi,\, \Upsilon)\, \mathrm{d}t - \varepsilon\int_0^T (\nabla\Theta,\, \nabla\Upsilon)\, \mathrm{d}t + \frac{1}{\varepsilon}\int_0^T (f(\Theta),\, \Upsilon)\, \mathrm{d}t = 0, \tag{83e}$$

$$\int_0^T (\nabla\cdot\mathbf{U},\, Q)\, \mathrm{d}t = 0. \tag{83f}$$

6.7 Main Convergence Theorem

Theorem 6.7 (Convergence to a weak solution)**.** *Under the conditions of Theorems 4.5 and 5.5, with $\mathbf{h}_a \in L^2(0,T;[H^1(\Omega)]^d)$, and assuming Assumption 6.1 (uniform $L^4_{t,x}$ bound on $\bar{\mathbf{M}}_{h,\tau}$), let $\{(\bar{\mathbf{U}}_{h,\tau}, \bar{\mathbf{M}}_{h,\tau}, \bar{\Phi}_{h,\tau}, \bar{\Theta}_{h,\tau}, \bar{\Psi}_{h,\tau}, \bar{P}_{h,\tau})\}$ be the sequence of piecewise-constant extensions of solutions of the scheme as $h, \tau \to 0$. Then there exists a subsequence along which the weak and strong convergences of Proposition 6.5 hold, and the limit tuple $(\mathbf{U}, \mathbf{M}, \Phi, \Theta, \Psi, P)$ is a weak solution in the sense of (83).*

The limit inherits the continuous energy inequality:

$$\mathcal{E}(t) + \int_0^t \mathcal{D}(s)\, \mathrm{d}s + 2\mu_0\beta\int_0^t \|\mathbf{M}\times\mathbf{H}\|^2_{L^2(\Omega)}\, \mathrm{d}s \;\le\; \mathcal{E}(0) + \int_0^t \mathcal{F}(s)\, \mathrm{d}s \quad \textit{for a.e. } t \in [0,T]. \tag{84}$$

Proof. Step 1: Weak and strong limits. By Propositions 6.2 and 6.5, along a subsequence all fields converge as stated.

Step 2: Passing to the limit in the scheme. Fix any smooth compactly supported test tuple as in (83). Write each scheme equation in integral form using the piecewise-constant extensions, multiply by τ, and sum over k. The resulting integrals over $[0,T]$ converge to the corresponding terms in (83) as follows.

Linear terms (time-derivative, viscosity, relaxation, Poisson, incompressibility, pressure, linear CH terms): pass to the limit by the weak convergences (70)–(74). The time-derivative

terms: after summation by parts, the discrete time-derivative term $\sum_k \left(\delta \mathbf{U}^k/\tau,\, \mathbf{V}^k\right) \tau$ converges to $-\int_0^T (\mathbf{U},\, \partial_t \mathbf{V})\, \mathrm{d}t$ using (75) and the smoothness of $\mathbf{V}$.

Nonlinear terms (NS advection, CH advection, T1, T2, T3, Kelvin, CH force): pass to the limit by Lemma 6.6 and the pointwise products established there. For the CH force: by (79) and the $L^2(H^1)$ bound on Ψ, the term $\left(\Theta^{k-1}\nabla\Psi^k,\, \mathbf{V}\right)$ converges to $(\Theta\nabla\Psi,\, \mathbf{V})$.

DG face terms: as shown in Section 6.5, the interior face contributions in B_h^m vanish as $h \to 0$ for smooth test functions, so the DG transport form converges to the continuous advection form.

Step 3: Incompressibility. Passing to the limit in $\left(\nabla \cdot \mathbf{U}^k,\, Q\right) = 0$ using (70) gives (83f).

Step 4: Pressure identification. By Lemma 6.4, $\bar{P}_{h,\tau}$ is bounded in $L^2(0,T;L^2(\Omega))$ and converges weakly to a limit P along a subsequence. Passing to the limit in the NS equation identifies P as the pressure in (83a).

Step 5: Energy inequality. Sum (27) from $k = 1$ to $k = \lfloor t/\tau \rfloor$ to get a discrete energy inequality on $[0, t]$. Pass to the limit using the weak lower semicontinuity of squared L^2 norms (for the energy and dissipation terms) and the strong convergence (75), (76) (for the β-dissipation term). This gives (84). □

Remark 6.8 (Scope and extensions)**.** Theorem 6.7 establishes convergence of a subsequence to a weak solution. Uniqueness of the weak solution (and hence convergence of the full sequence) and quantitative error estimates require additional regularity of the solution (e.g., $\mathbf{U} \in L^2(0,T;[H^2(\Omega)]^d)$, $\mathbf{M} \in L^2(0,T;[H^1(\Omega)]^d)$) and are the subject of future work. The result extends the analogous statement in [2] to the full Shliomis model including the T1, T2, and T3 terms of [3].

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